

Star Vetter - confirmed eclipsing binary candidates

TIC	Period (d)	Depth	BLS SNR	Pixel SNR	Offset	Tier
437150434	5.2853	15.003%	1081	85.33	1"	Clean
237115819	14.7921	8.883%	507	27.82	4"	Clean
66732476	3.1087	9.814%	370	26.93	8"	Clean
29495157	5.1474	8.579%	333	52.62	5"	Alias
119459681	5.9442	15.977%	311	68.15	2"	Clean
411434067	1.3536	4.289%	193	33.70	4"	Alias
330901906	3.7321	0.498%	60	87.24	4"	Thin margin
303470667	5.4319	2.220%	57	5.02	9"	Marginal
273158289	1.1519	1.255%	41	17.24	2"	Thin margin
111149941	5.2401	0.045%	10	9.73	21"	Ambiguous
251898081	27.3089	18.630%	5535	464.16	3"	Alias
467214801	0.7609	0.950%	2134	21.36	0"	Alias
399770041	1.5802	4.291%	489	36.92	1"	Clean
349701853	30.3427	28.445%	343	51.05	6"	Clean
233459038	25.3384	8.315%	237	84.46	5"	Alias
281310348	1.7702	7.273%	180	57.87	2"	Clean
318887646	14.2817	3.605%	150	6.94	13"	Clean
444465823	18.5133	3.476%	147	25.18	1"	Alias
380943251	0.7209	7.667%	101	15.26	3"	Clean
66716482	8.0283	3.329%	65	21.51	7"	Clean
309656476	28.9204	0.050%	25	13.80	14"	Clean
229494833	15.3839	0.037%	15	5.19	12"	Alias
44533527	14.5952	0.101%	12	7.33	16"	Clean
220423668	3.6511	4.525%	755	17.67	6"	Alias
365998904	1.2494	2.398%	375	15.52	3"	Clean
233452636	1.9063	5.467%	202	8.94	2"	Thin margin
437171483	1.4735	4.947%	923	73.37	3"	Clean
411457490	2.4324	5.531%	431	58.00	1"	Alias
295373509	5.4219	8.026%	247	22.12	2"	Clean
374158907	14.6563	4.974%	179	21.40	2"	Alias

TIC 437150434

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 5.285343 d
Epoch (t0) 2230.8422 BTJD
Duration 0.1700 d
Depth 15.003 % (150,033 ppm)
SNR 1081.0

INDEPENDENT CHECKS (f3i gates)

Eclipses observed 4 (gate ≥ 3)
Robust depth ratio 1.18 (gate ≥ 0.50)
Robust depth sigma 891.6 (gate ≥ 7.0)
Odd/even z-score 1.10 (consistent)
Secondary eclipse not detected

CATALOG CROSS-MATCH (novelty)

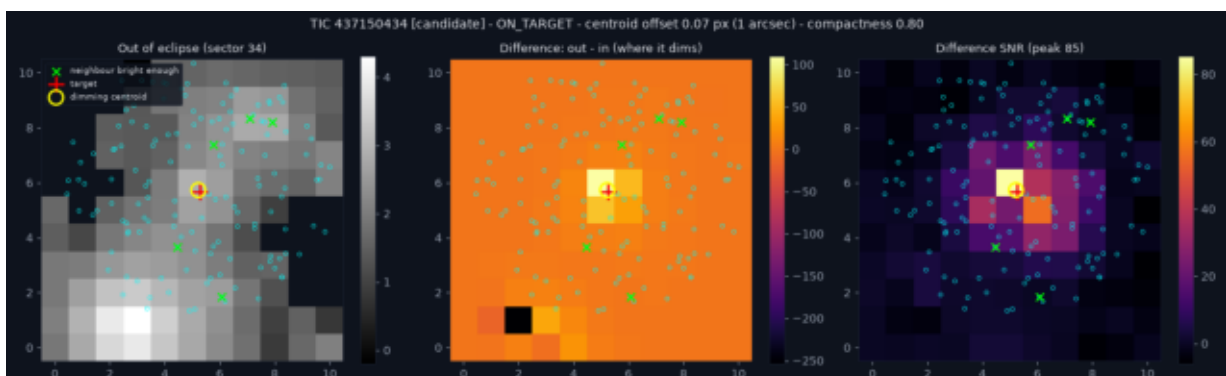
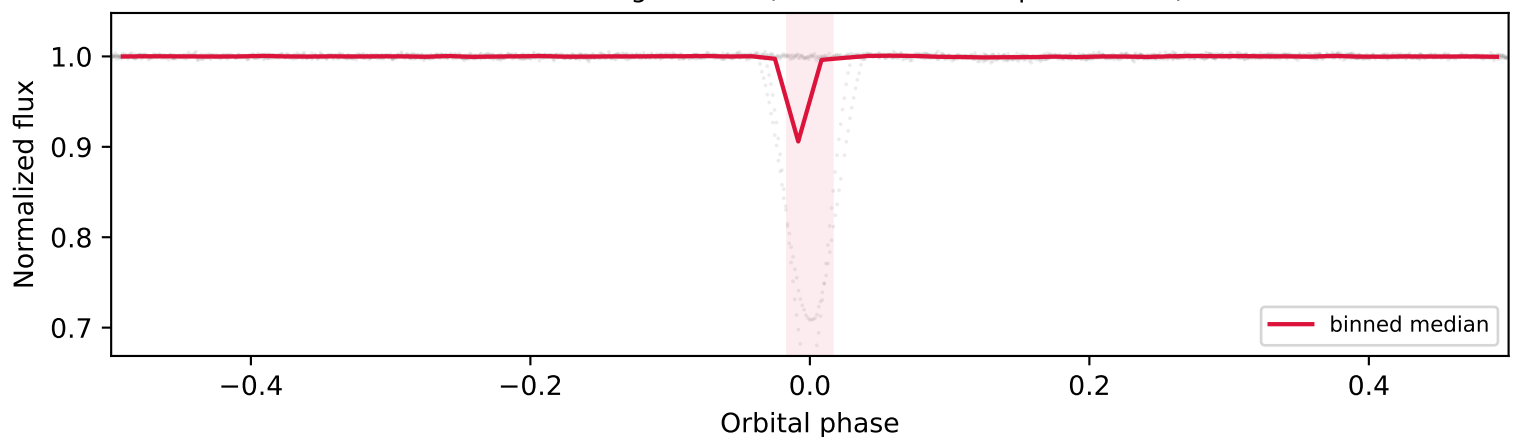
VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 34 light-curve dip in this sector: 645.4 sigma t0 refined by -0.0075 phase
Difference-image peak SNR 85.33 (gate ≥ 5.0) Compactness 0.80 (gate ≥ 0.50)
Dimming centroid offset 0.07 px (1 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 437150402
1.75 px away, dTmag +1.46 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.

TIC 237115819

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 14.792059 d
Epoch (t0) 1683.4391 BTJD
Duration 0.0840 d
Depth 8.883 % (88,832 ppm)
SNR 507.0

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

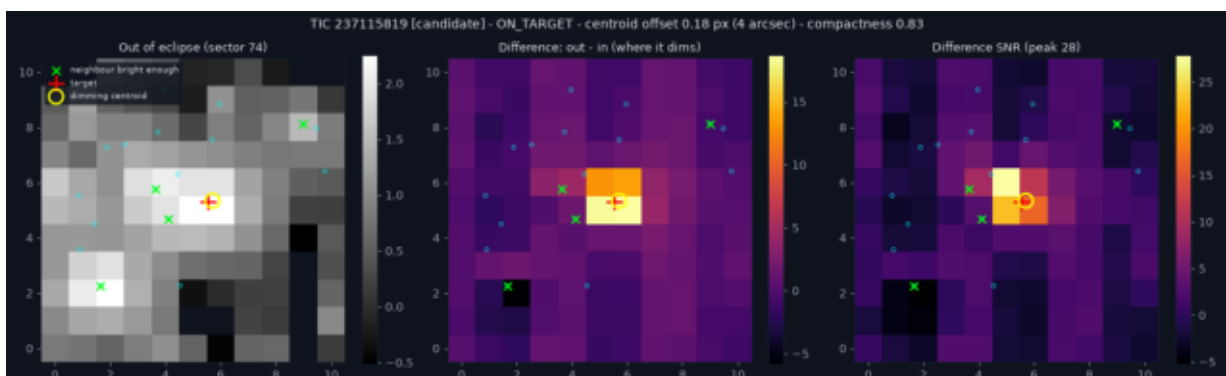
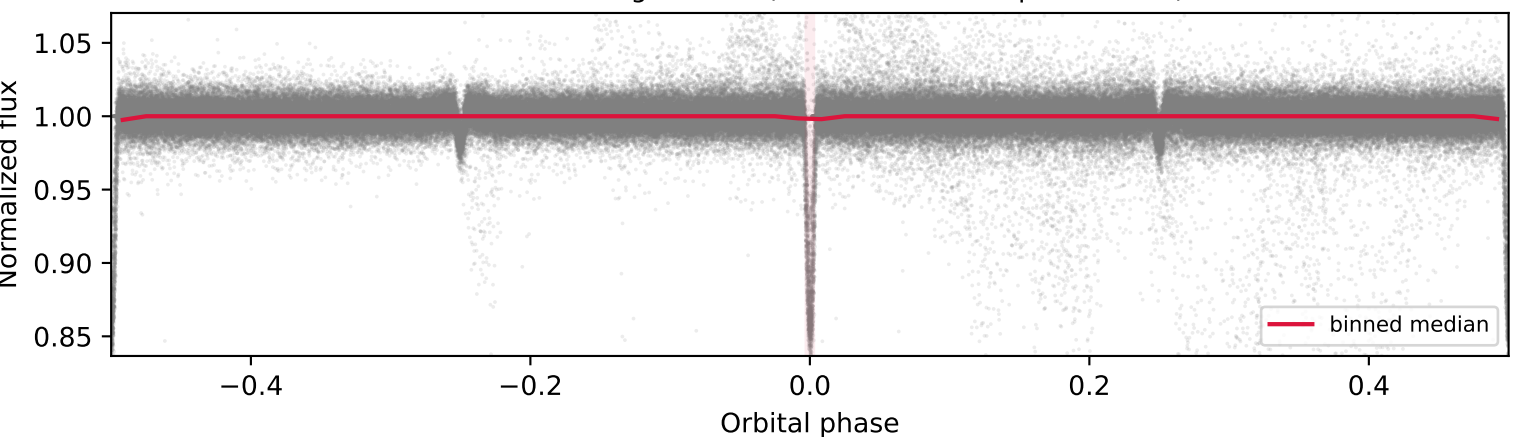
Eclipses observed 75 (gate ≥ 3)
Robust depth ratio 1.23 (gate ≥ 0.50)
Robust depth sigma 414.0 (gate ≥ 7.0)
Odd/even z-score 1.61 (consistent)
Secondary eclipse detected, phase -0.49, 7.9 sigma

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 74 light-curve dip in this sector: 51.1 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 27.82 (gate ≥ 5.0) Compactness 0.83 (gate ≥ 0.50)
Dimming centroid offset 0.18 px (4 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 237115821
1.72 px away, dTmag +2.50 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.

TIC 66732476

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 3.108737 d
Epoch (t0) 2553.3754 BTJD
Duration 0.0840 d
Depth 9.814 % (98,141 ppm)
SNR 369.7

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

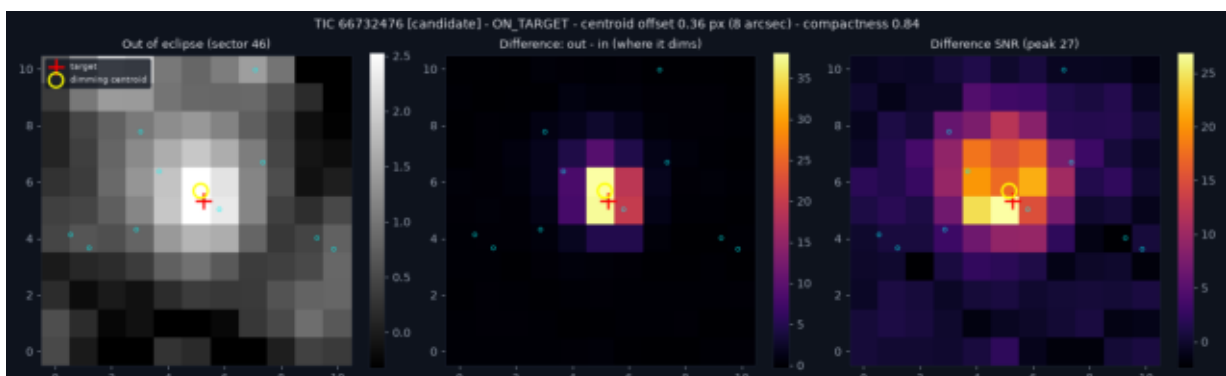
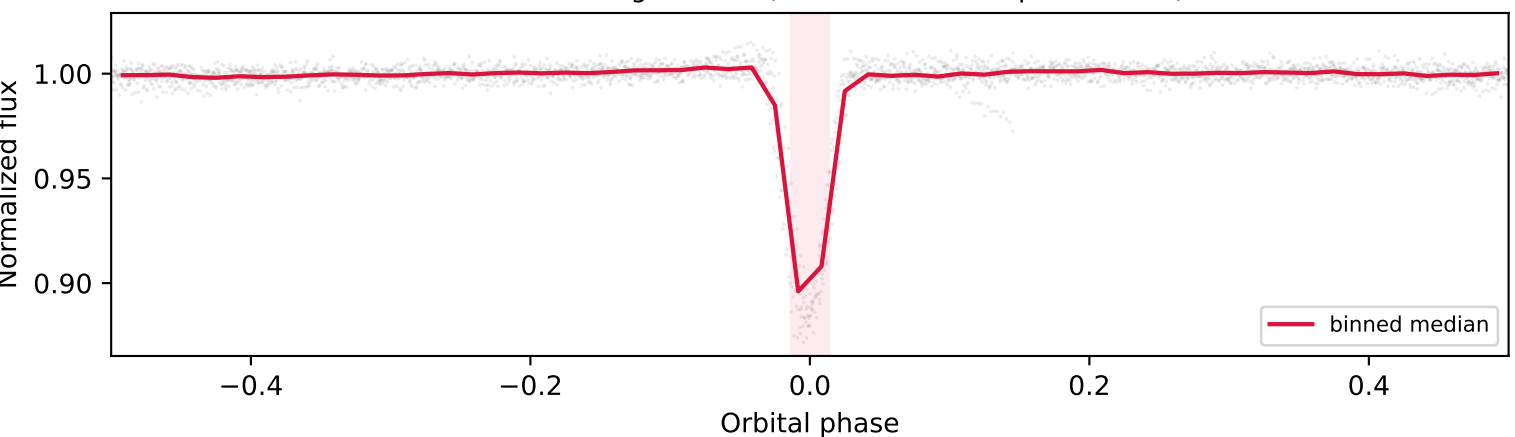
Eclipses observed 8 (gate ≥ 3)
Robust depth ratio 1.06 (gate ≥ 0.50)
Robust depth sigma 247.6 (gate ≥ 7.0)
Odd/even z-score 2.60 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 46 light-curve dip in this sector: 234.7 sigma t0 refined by -0.0050 phase
Difference-image peak SNR 26.93 (gate ≥ 5.0) Compactness 0.84 (gate ≥ 0.50)
Dimming centroid offset 0.36 px (8 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: none within the cutout

Phase-folded light curve (shaded = BLS eclipse window)



TIC 29495157

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 5.147449 d (ALIASED, see below)
-> corrected period: 10.294897 d
Epoch (t0) 1328.1391 BTJD
Duration 0.1700 d
Depth 8.579 % (85,786 ppm)
SNR 333.2

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

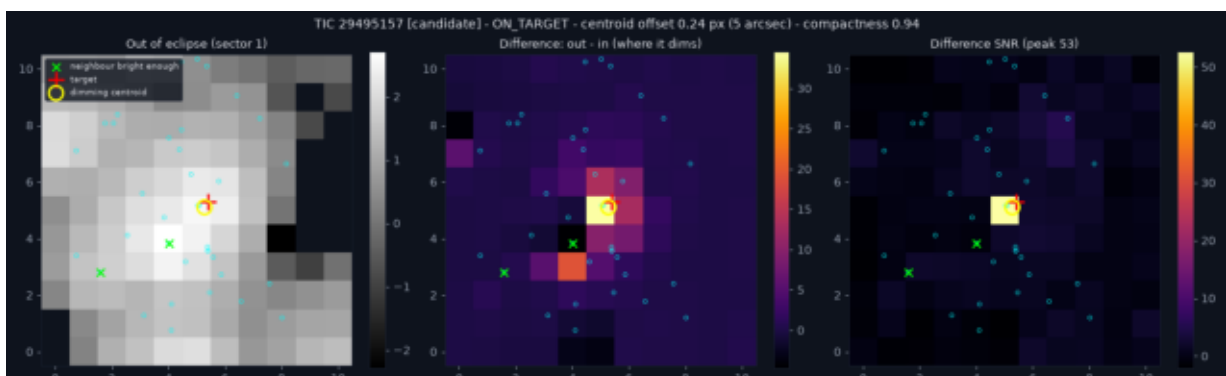
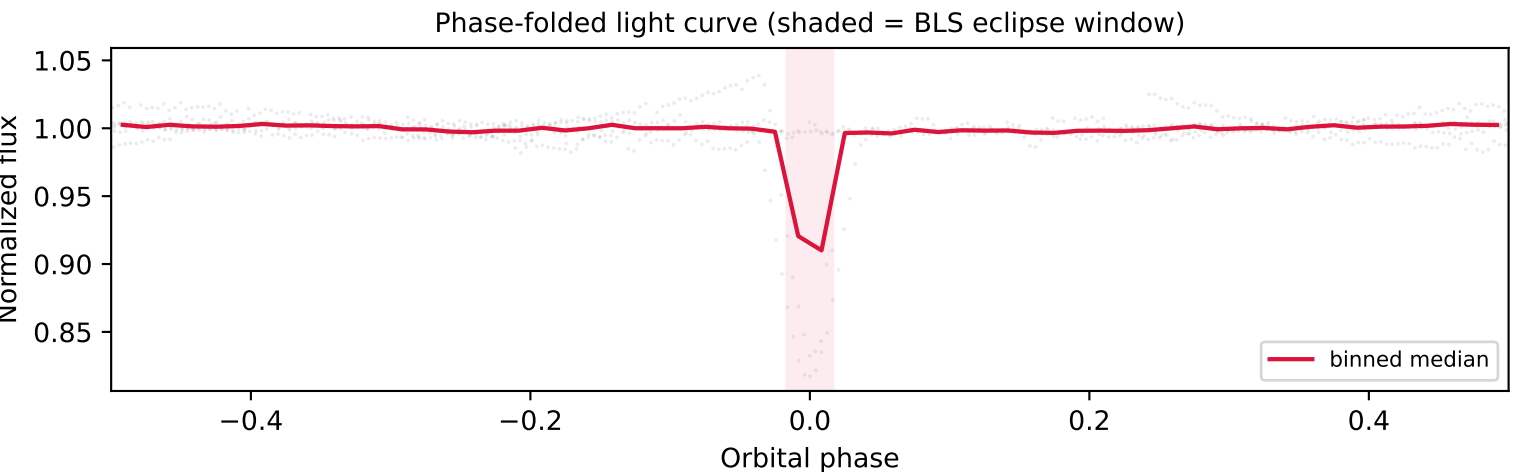
INDEPENDENT CHECKS (f3i gates)

Eclipses observed 4 (gate ≥ 3)
Robust depth ratio 1.05 (gate ≥ 0.50)
Robust depth sigma 86.0 (gate ≥ 7.0)
Odd/even z-score 20.17 (ALIAS FLAGGED)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 1 light-curve dip in this sector: 94.0 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 52.62 (gate ≥ 5.0) Compactness 0.94 (gate ≥ 0.50)
Dimming centroid offset 0.24 px (5 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 29495166
1.79 px away, dTmag -0.48 - farther from the dimming than the target



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 20.17 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 10.294897 d, in any write-up.

TIC 119459681

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 5.944205 d
Epoch (t0) 1471.4023 BTJD
Duration 0.2440 d
Depth 15.977 % (159,771 ppm)
SNR 310.6

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

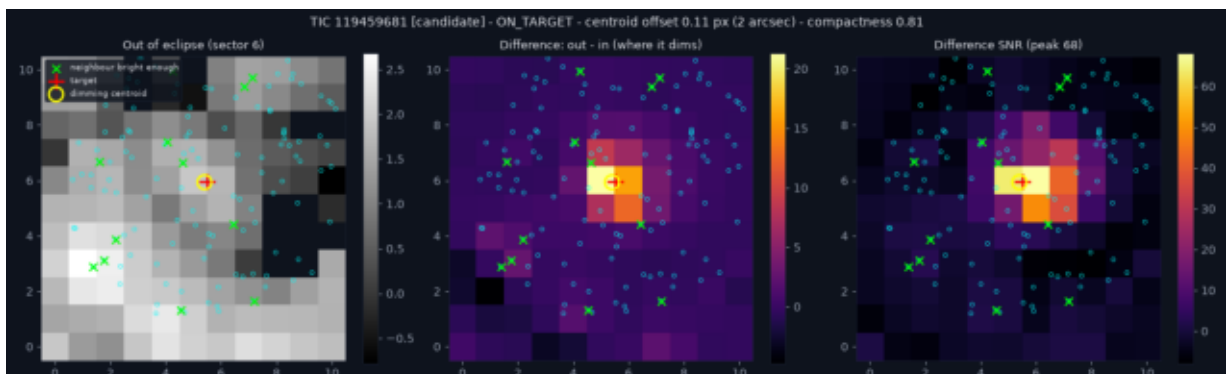
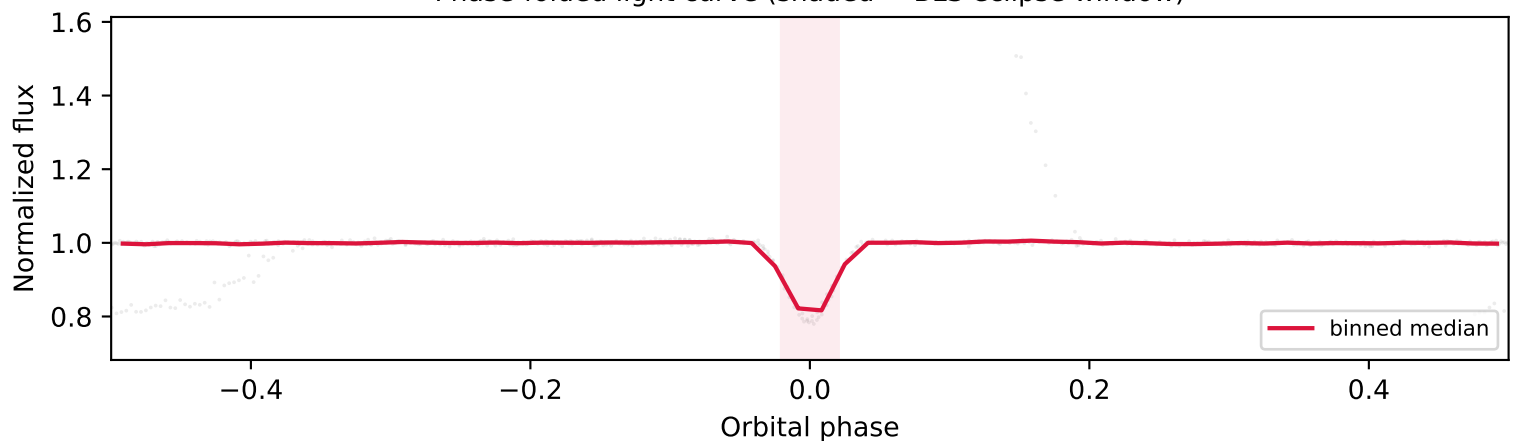
Eclipses observed 3 (gate ≥ 3)
Robust depth ratio 1.09 (gate ≥ 0.50)
Robust depth sigma 199.8 (gate ≥ 7.0)
Odd/even z-score 1.19 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 6 light-curve dip in this sector: 183.3 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 68.15 (gate ≥ 5.0) Compactness 0.81 (gate ≥ 0.50)
Dimming centroid offset 0.11 px (2 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 119459668
1.05 px away, dTmag +1.18 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.

TIC 411434067

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 1.353604 d (ALIASED, see below)
-> corrected period: 2.707209 d
Epoch (t0) 2825.8292 BTJD
Duration 0.1200 d
Depth 4.289 % (42,887 ppm)
SNR 192.9

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

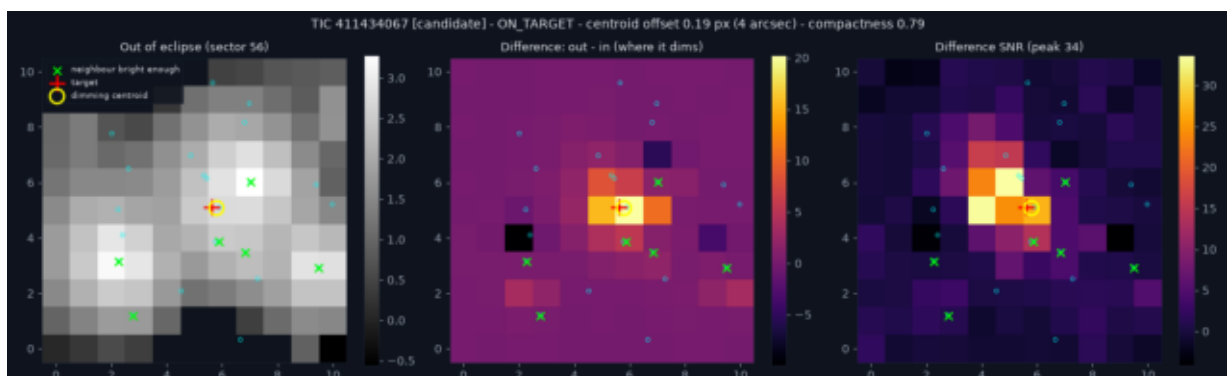
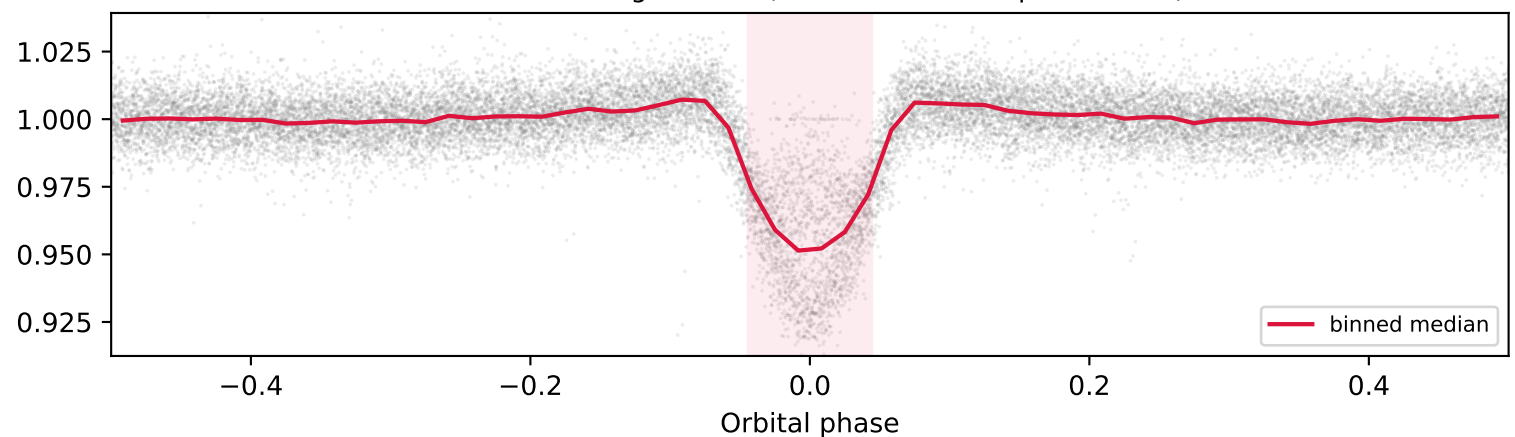
Eclipses observed 39 (gate ≥ 3)
Robust depth ratio 0.92 (gate ≥ 0.50)
Robust depth sigma 162.8 (gate ≥ 7.0)
Odd/even z-score 9.80 (ALIAS FLAGGED)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 56 light-curve dip in this sector: 130.8 sigma t0 refined by -0.0025 phase
Difference-image peak SNR 33.70 (gate ≥ 5.0) Compactness 0.79 (gate ≥ 0.50)
Dimming centroid offset 0.19 px (4 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 411429819
1.21 px away, dTmag +1.19 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 9.80 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 2.707209 d, in any write-up.

TIC 330901906

Eclipsing binary candidate dossier - Star Vetter

THIN MARGIN

BLS EPHEMERIS		CATALOG CROSS-MATCH (novelty)	
Period	3.732068 d	VSX	no match
Epoch (t0)	1740.9150 BTJD	ASAS-SN	no match
Duration	0.2440 d	Gaia eclipsing-binary table	no match
Depth	0.498 % (4,983 ppm)	Gaia variable classification	no match
SNR	60.1	TESS EB catalog (Prsa+2022)	no match
INDEPENDENT CHECKS (f3i gates)		ExoFOP TOI list	no match
Eclipses observed	29 (gate >= 3)	Verdict:	NOVEL
Robust depth ratio	1.19 (gate >= 0.50)		
Robust depth sigma	58.6 (gate >= 7.0)		
Odd/even z-score	0.36 (consistent)		
Secondary eclipse	not detected		

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

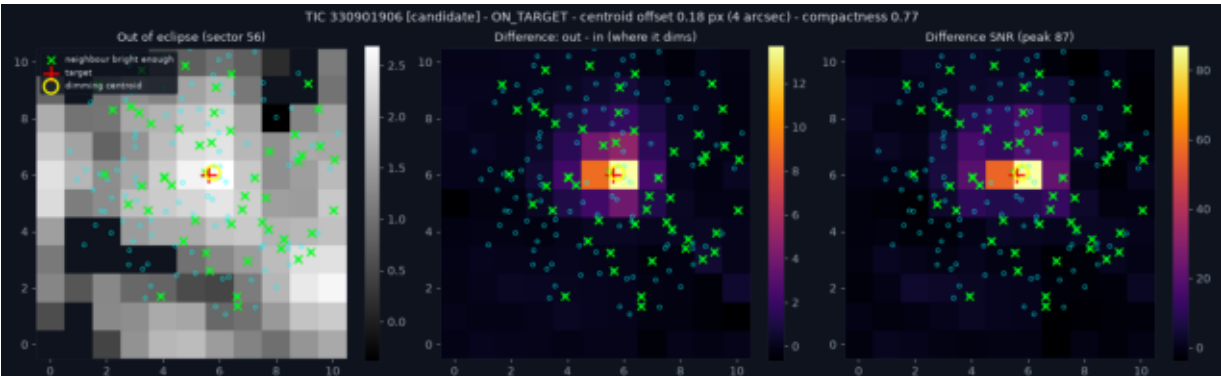
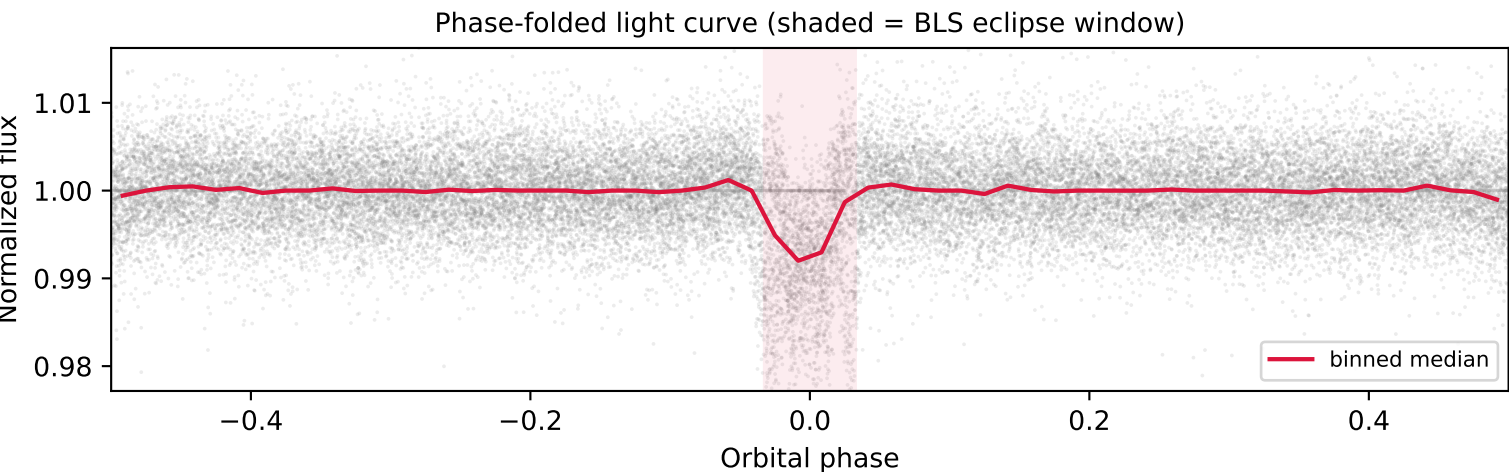
Sector 56 light-curve dip in this sector: 38.2 sigma t0 refined by 0.0075 phase

Difference-image peak SNR 87.24 (gate >= 5.0) Compactness 0.77 (gate >= 0.50)

Dimming centroid offset 0.18 px (4 arcsec) (gate <= 1.0 px, i.e. on the target)

Nearest bright-enough neighbour in cutout: TIC 2009125529

0.25 px away, dTmag +4.30 - farther from the dimming than the target



Caveat (THIN MARGIN): The ON_TARGET call rests on a neighbour-competition margin of just 0.066 px (dimming centroid 0.182 px from target vs 0.248 px from the nearest bright-enough neighbour) - inside diag_pixel.py's own stated centroid systematic of 0.10-0.3 px. TESS's resolution cannot confidently separate this star from that neighbour.

TIC 303470667

Eclipsing binary candidate dossier - Star Vetter

MARGINAL

BLS EPHEMERIS		CATALOG CROSS-MATCH (novelty)	
Period	5.431910 d	VSX	no match
Epoch (t0)	1657.9950 BTJD	ASAS-SN	no match
Duration	0.1200 d	Gaia eclipsing-binary table	no match
Depth	2.220 % (22,198 ppm)	Gaia variable classification	no match
SNR	57.0	TESS EB catalog (Prsa+2022)	no match
INDEPENDENT CHECKS (f3i gates)		ExoFOP TOI list	no match
Eclipses observed	4 (gate >= 3)	Verdict:	NOVEL
Robust depth ratio	1.03 (gate >= 0.50)		
Robust depth sigma	38.6 (gate >= 7.0)		
Odd/even z-score	2.61 (consistent)		
Secondary eclipse	not detected		

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

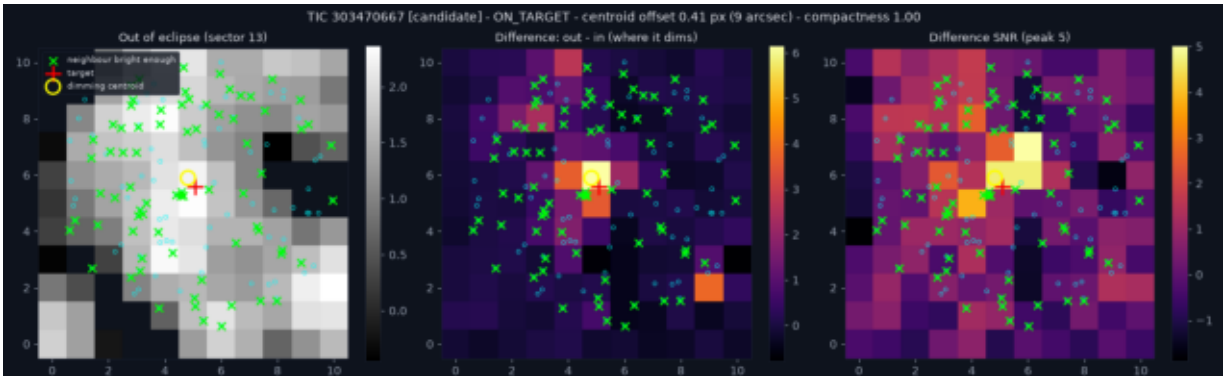
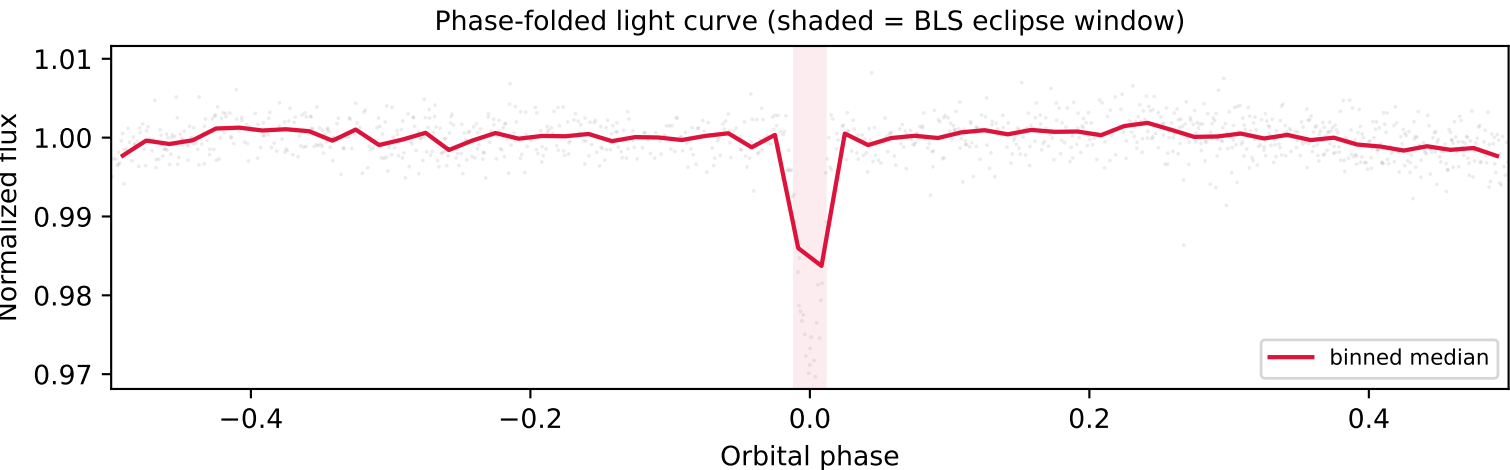
Sector 13 light-curve dip in this sector: 33.5 sigma t0 refined by 0.0025 phase

Difference-image peak SNR 5.02 (gate >= 5.0) Compactness 1.00 (gate >= 0.50)

Dimming centroid offset 0.41 px (9 arcsec) (gate <= 1.0 px, i.e. on the target)

Nearest bright-enough neighbour in cutout: TIC 1689765373

0.54 px away, dTmag +3.73 - farther from the dimming than the target



Caveat (MARGINAL): Pixel-check SNR is right at the gate (5.02 vs >= 5.0). A 150-trial false-alarm check on this star's own data found only a 3.3% chance of reaching this SNR at a random phase - the detection looks real, just faint. Weakest SNR margin of the confirmed set.

TIC 273158289

Eclipsing binary candidate dossier - Star Vetter

THIN MARGIN

BLS EPHEMERIS		CATALOG CROSS-MATCH (novelty)	
Period	1.151883 d	VSX	no match
Epoch (t0)	1597.9084 BTJD	ASAS-SN	no match
Duration	0.0840 d	Gaia eclipsing-binary table	no match
Depth	1.255 % (12,546 ppm)	Gaia variable classification	no match
SNR	41.3	TESS EB catalog (Prsa+2022)	no match
INDEPENDENT CHECKS (f3i gates)		ExoFOP TOI list	no match
Eclipses observed	19 (gate >= 3)	Verdict: NOVEL	
Robust depth ratio	1.03 (gate >= 0.50)		
Robust depth sigma	28.6 (gate >= 7.0)		
Odd/even z-score	1.42 (consistent)		
Secondary eclipse	not detected		

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

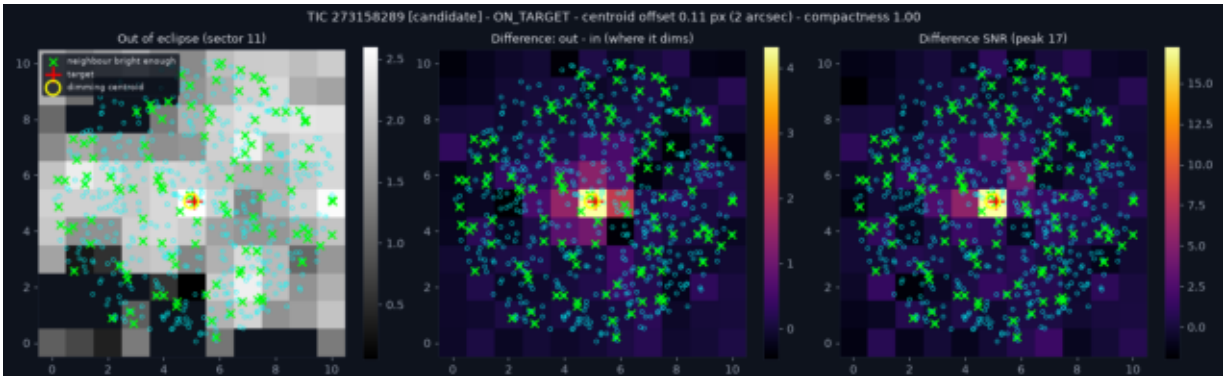
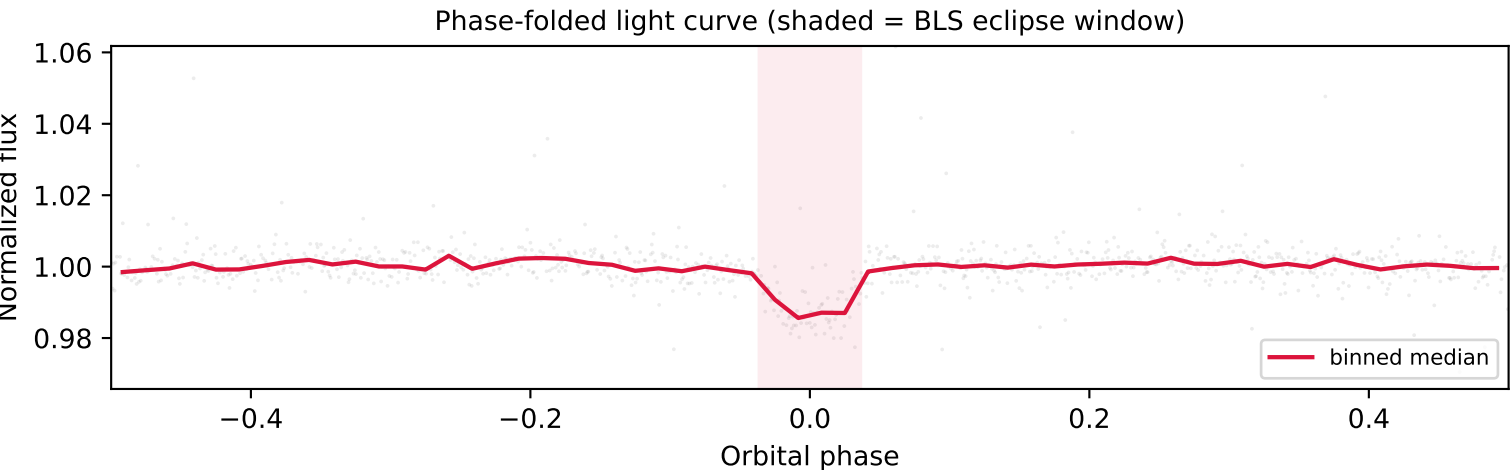
Sector 11 light-curve dip in this sector: 27.4 sigma t0 refined by 0.0050 phase

Difference-image peak SNR 17.24 (gate >= 5.0) Compactness 1.00 (gate >= 0.50)

Dimming centroid offset 0.11 px (2 arcsec) (gate <= 1.0 px, i.e. on the target)

Nearest bright-enough neighbour in cutout: TIC 514580937

0.12 px away, dTmag +0.71 - farther from the dimming than the target



Caveat (THIN MARGIN): The ON_TARGET call rests on a neighbour-competition margin of just 0.006 px (dimming centroid 0.113 px from target vs 0.119 px from the nearest bright-enough neighbour) - inside diag_pixel.py's own stated centroid systematic of 0.10-0.3 px. TESS's resolution cannot confidently separate this star from that neighbour.

TIC 111149941

Eclipsing binary candidate dossier - Star Vetter

AMBIGUOUS PHOTOMETRY

BLS EPHEMERIS		CATALOG CROSS-MATCH (novelty)	
Period	5.240092 d	VSX	no match
Epoch (t0)	2695.2575 BTJD	ASAS-SN	no match
Duration	0.3500 d	Gaia eclipsing-binary table	no match
Depth	0.045 % (455 ppm)	Gaia variable classification	no match
SNR	10.2	TESS EB catalog (Prsa+2022)	no match
INDEPENDENT CHECKS (f3i gates)		ExoFOP TOI list	no match
Eclipses observed	3 (gate >= 3)	Verdict:	NOVEL
Robust depth ratio	1.02 (gate >= 0.50)		
Robust depth sigma	8.5 (gate >= 7.0)		
Odd/even z-score	2.29 (consistent)		
Secondary eclipse	not detected		

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

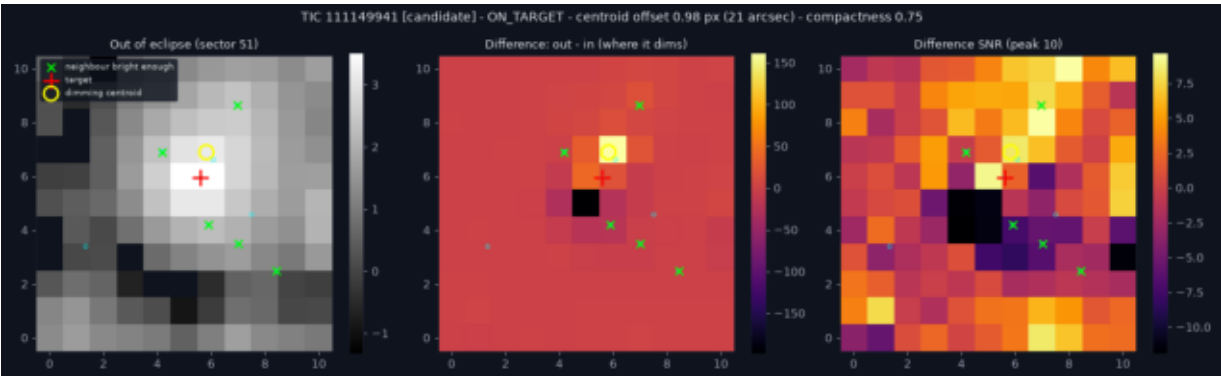
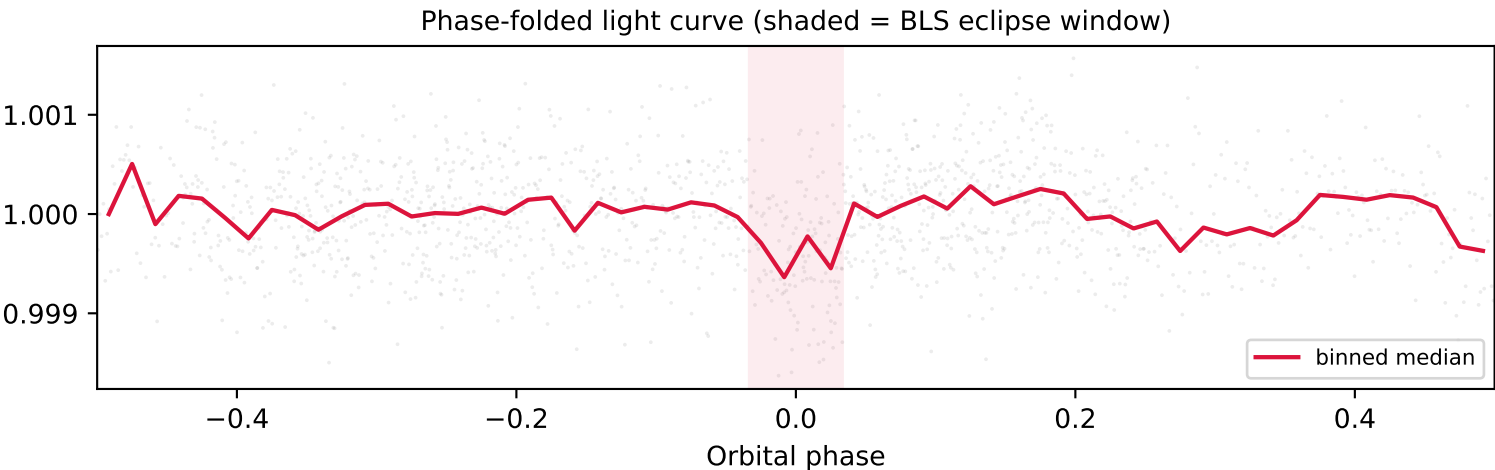
Sector 51 light-curve dip in this sector: 7.9 sigma t0 refined by 0.0100 phase

Difference-image peak SNR 9.73 (gate >= 5.0) Compactness 0.75 (gate >= 0.50)

Dimming centroid offset 0.98 px (21 arcsec) (gate <= 1.0 px, i.e. on the target)

Nearest bright-enough neighbour in cutout: TIC 1002359509

1.65 px away, dTmag +8.06 - farther from the dimming than the target



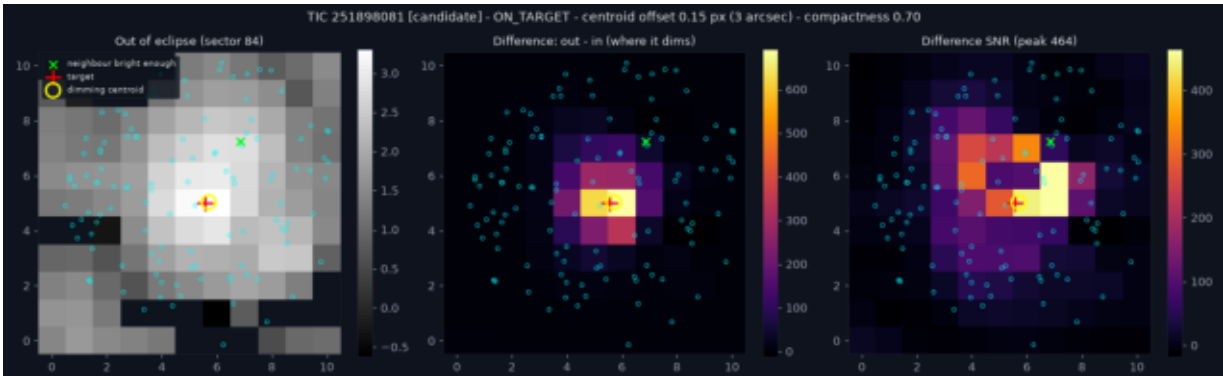
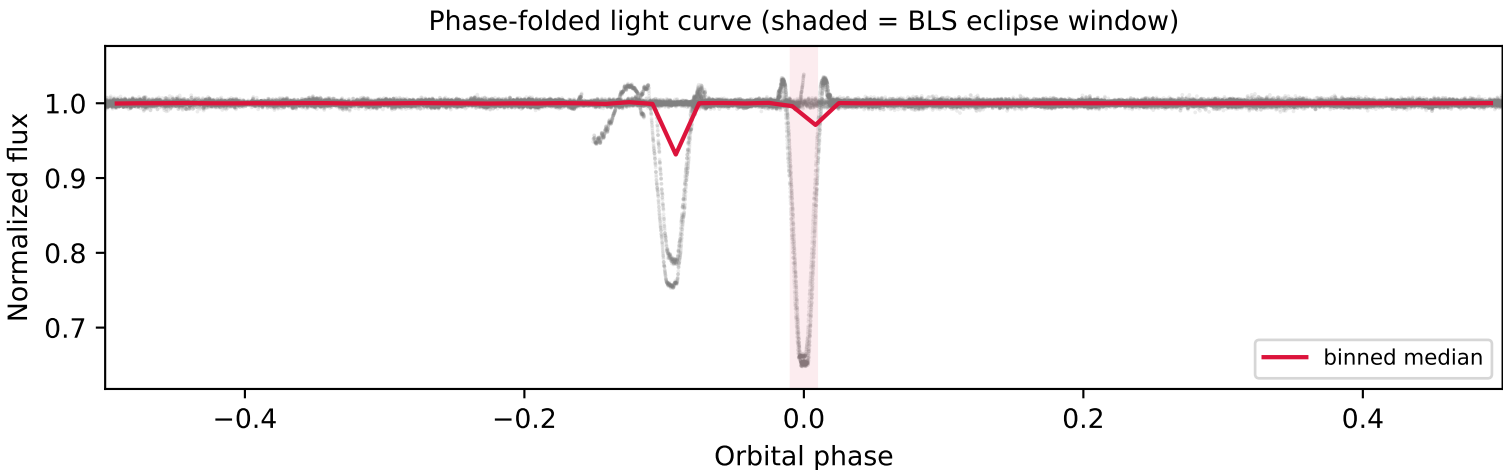
Caveat (AMBIGUOUS PHOTOMETRY): The difference-image detection is strong (peak SNR 9.7, compactness 0.75) but the star's own simple-aperture photometry shows no dip, and aperture/quality-mask sensitivity tests were inconsistent. Confirmed by pixel imaging; not independently confirmed by simple photometry.

TIC 251898081

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS		CATALOG CROSS-MATCH (novelty)	
Period	27.308942 d (ALIASED, see below)	VSX	no match
-> corrected period:	54.617884 d	ASAS-SN	no match
Epoch (t0)	2876.1224 BTJD	Gaia eclipsing-binary table	no match
Duration	0.5000 d	Gaia variable classification	no match
Depth	18.630 % (186,304 ppm)	TESS EB catalog (Prsa+2022)	no match
SNR	5534.5	ExoFOP TOI list	no match
INDEPENDENT CHECKS (f3i gates)		Verdict: NOVEL	
Eclipses observed	5 (gate >= 3)		
Robust depth ratio	1.07 (gate >= 0.50)		
Robust depth sigma	1819.7 (gate >= 7.0)		
Odd/even z-score	198.85 (ALIAS FLAGGED)		
Secondary eclipse	detected, phase -0.09, 9.9 sigma		
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)			
Sector 84	light-curve dip in this sector: 1459.8 sigma	t0 refined by 0.0025 phase	
Difference-image peak SNR	464.16 (gate >= 5.0)	Compactness	0.70 (gate >= 0.50)
Dimming centroid offset	0.15 px (3 arcsec) (gate <= 1.0 px, i.e. on the target)		
Nearest bright-enough neighbour in cutout: TIC 251898060			
2.48 px away, dTmag +1.74 - farther from the dimming than the target			



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 198.85 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 54.617884 d, in any write-up.

TIC 467214801

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS		CATALOG CROSS-MATCH (novelty)	
Period	0.760850 d (ALIASED, see below)	VSX	no match
-> corrected period:	1.521701 d	ASAS-SN	no match
Epoch (t0)	2825.3015 BTJD	Gaia eclipsing-binary table	no match
Duration	0.0840 d	Gaia variable classification	no match
Depth	0.950 % (9,505 ppm)	TESS EB catalog (Prsa+2022)	no match
SNR	2133.7	ExoFOP TOI list	no match
INDEPENDENT CHECKS (f3i gates)		Verdict: NOVEL	
Eclipses observed	329 (gate >= 3)		
Robust depth ratio	1.11 (gate >= 0.50)		
Robust depth sigma	104.6 (gate >= 7.0)		
Odd/even z-score	5.40 (ALIAS FLAGGED)		
Secondary eclipse	not detected		

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

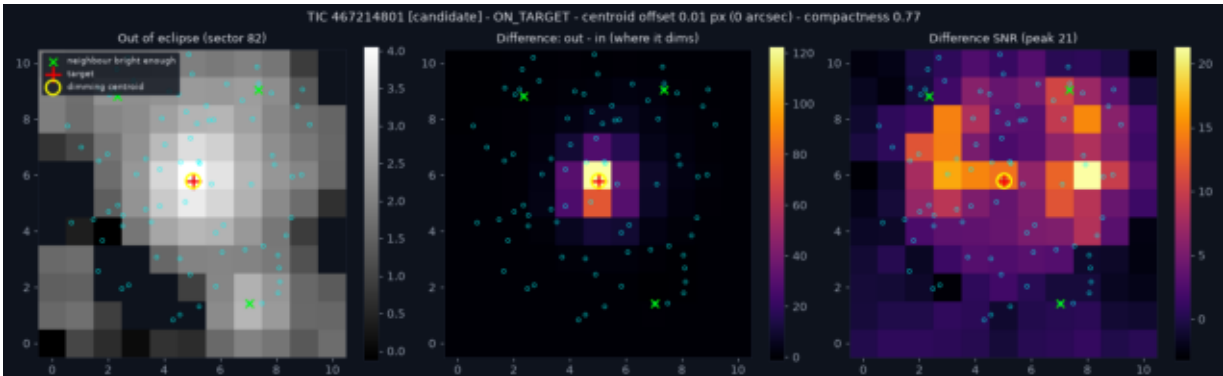
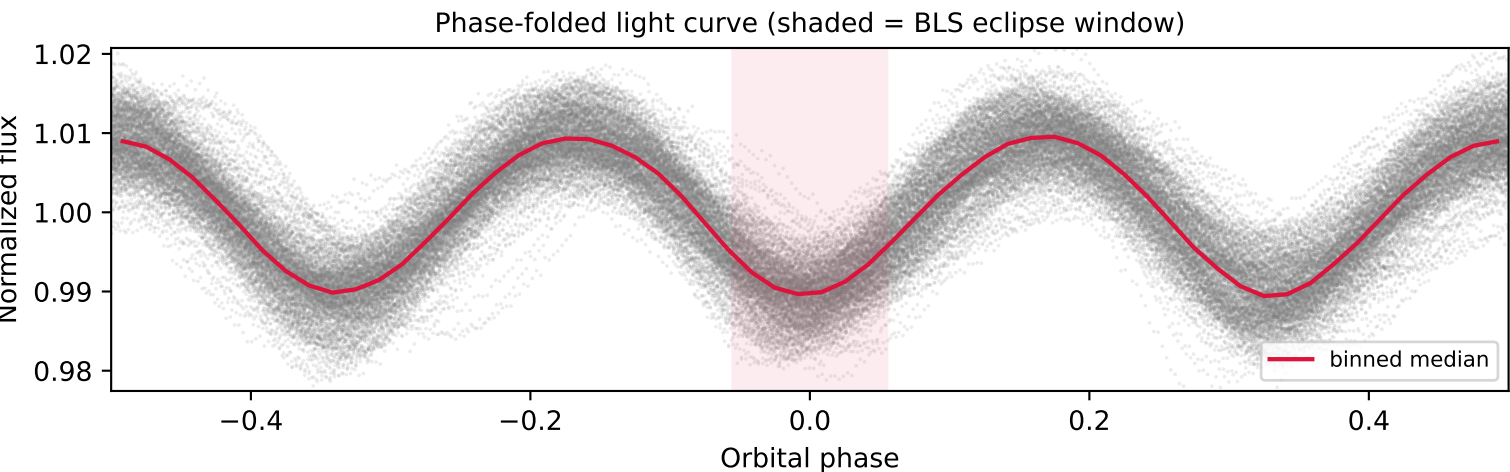
Sector 82 light-curve dip in this sector: 25.9 sigma t0 refined by -0.0100 phase

Difference-image peak SNR 21.36 (gate >= 5.0) Compactness 0.77 (gate >= 0.50)

Dimming centroid offset 0.01 px (0 arcsec) (gate <= 1.0 px, i.e. on the target)

Nearest bright-enough neighbour in cutout: TIC 467214771

4.01 px away, dTmag +4.53 - farther from the dimming than the target



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 5.40 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 1.521701 d, in any write-up.

TIC 399770041

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 1.580189 d
Epoch (t0) 1569.9907 BTJD
Duration 0.0840 d
Depth 4.291 % (42,906 ppm)
SNR 488.6

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

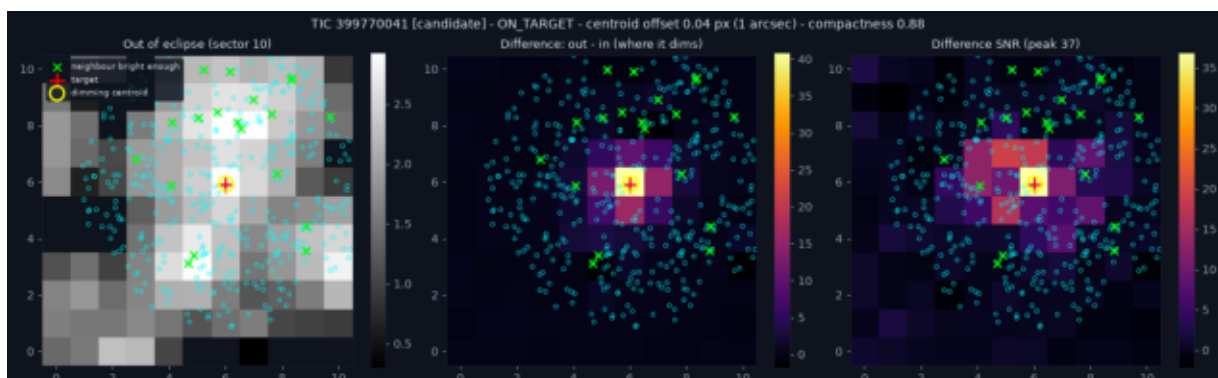
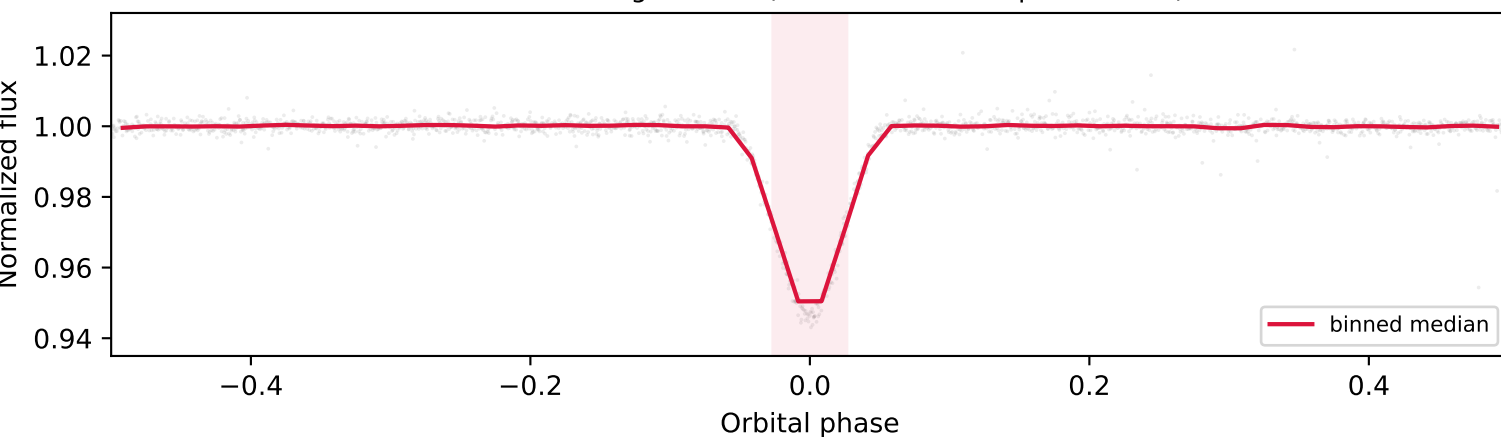
Eclipses observed 30 (gate ≥ 3)
Robust depth ratio 1.04 (gate ≥ 0.50)
Robust depth sigma 283.6 (gate ≥ 7.0)
Odd/even z-score 0.86 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 10 light-curve dip in this sector: 202.4 sigma t0 refined by -0.0075 phase
Difference-image peak SNR 36.92 (gate ≥ 5.0) Compactness 0.88 (gate ≥ 0.50)
Dimming centroid offset 0.04 px (1 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 399770066
1.83 px away, dTmag +2.10 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



TIC 349701853

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS		CATALOG CROSS-MATCH (novelty)	
Period	30.342709 d	VSX	no match
Epoch (t0)	2395.4286 BTJD	ASAS-SN	no match
Duration	0.1200 d	Gaia eclipsing-binary table	no match
Depth	28.445 % (284,452 ppm)	Gaia variable classification	no match
SNR	343.5	TESS EB catalog (Prsa+2022)	no match
INDEPENDENT CHECKS (f3i gates)		ExoFOP TOI list	no match
Eclipses observed	4 (gate >= 3)	Verdict:	NOVEL
Robust depth ratio	1.03 (gate >= 0.50)		
Robust depth sigma	252.5 (gate >= 7.0)		
Odd/even z-score	0.87 (consistent)		
Secondary eclipse	detected, phase 0.33, 10.0 sigma		

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

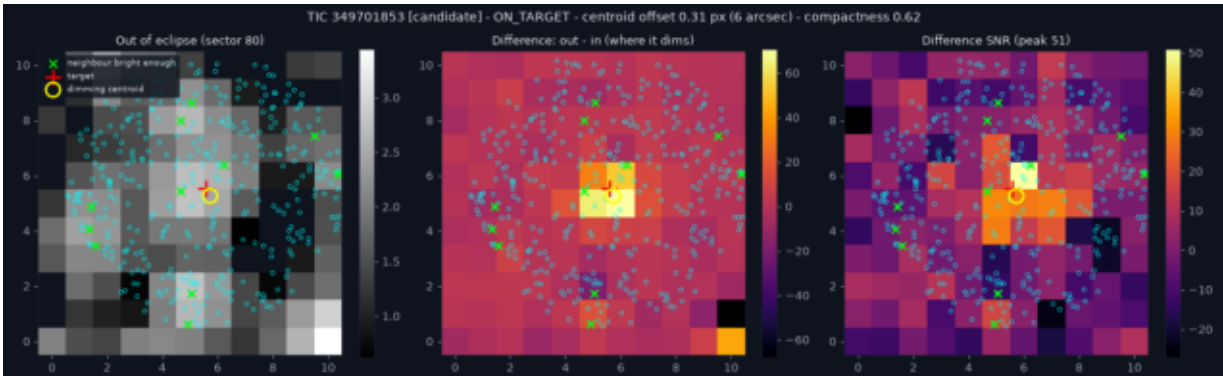
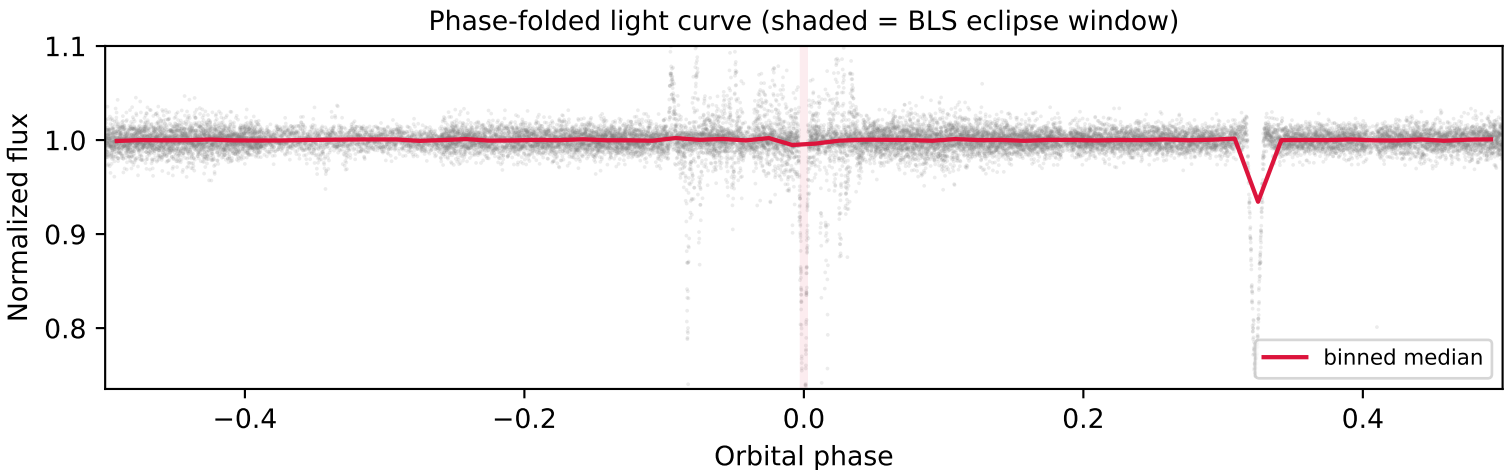
Sector 80 light-curve dip in this sector: 149.7 sigma t0 refined by 0.0000 phase

Difference-image peak SNR 51.05 (gate >= 5.0) Compactness 0.62 (gate >= 0.50)

Dimming centroid offset 0.31 px (6 arcsec) (gate <= 1.0 px, i.e. on the target)

Nearest bright-enough neighbour in cutout: TIC 349701859

1.07 px away, dTmag -0.58 - farther from the dimming than the target

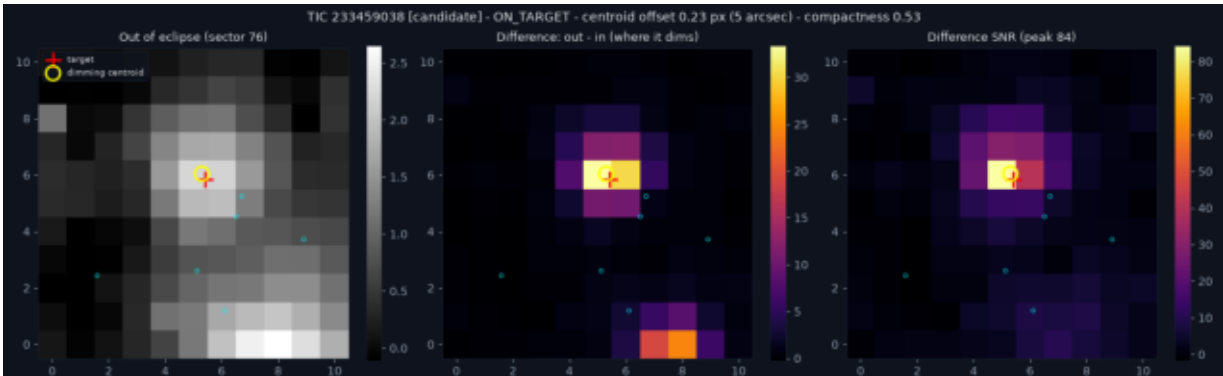
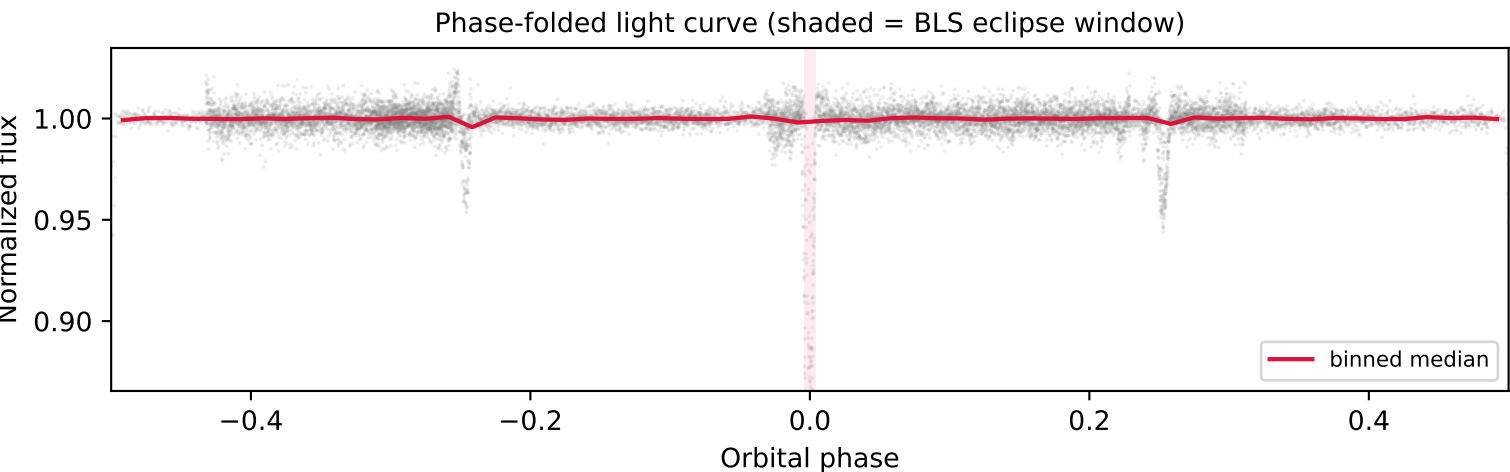


TIC 233459038

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS		CATALOG CROSS-MATCH (novelty)	
Period	25.338439 d (ALIASED, see below)		
-> corrected period: 50.676878 d		VSX	no match
Epoch (t0)	1728.5852 BTJD	ASAS-SN	no match
Duration	0.1700 d	Gaia eclipsing-binary table	no match
Depth	8.315 % (83,146 ppm)	Gaia variable classification	no match
SNR	236.6	TESS EB catalog (Prsa+2022)	no match
		ExoFOP TOI list	no match
INDEPENDENT CHECKS (f3i gates)			
Eclipses observed	6 (gate >= 3)	Verdict: NOVEL	
Robust depth ratio	1.37 (gate >= 0.50)		
Robust depth sigma	274.3 (gate >= 7.0)		
Odd/even z-score	89.17 (ALIAS FLAGGED)		
Secondary eclipse	detected, phase 0.26, 8.3 sigma		
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)			
Sector 76	light-curve dip in this sector: 169.1 sigma	t0 refined by 0.0000 phase	
Difference-image peak SNR	84.46 (gate >= 5.0)	Compactness 0.53 (gate >= 0.50)	
Dimming centroid offset	0.23 px (5 arcsec) (gate <= 1.0 px, i.e. on the target)		
Nearest bright-enough neighbour in cutout: none within the cutout			



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 89.17 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 50.676878 d, in any write-up.

TIC 281310348

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 1.770214 d
Epoch (t0) 1469.3509 BTJD
Duration 0.1200 d
Depth 7.273 % (72,730 ppm)
SNR 179.7

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

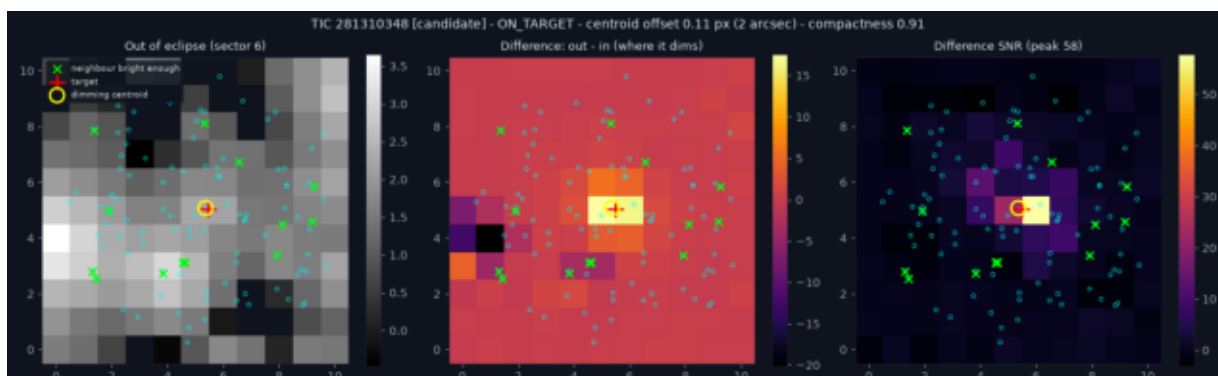
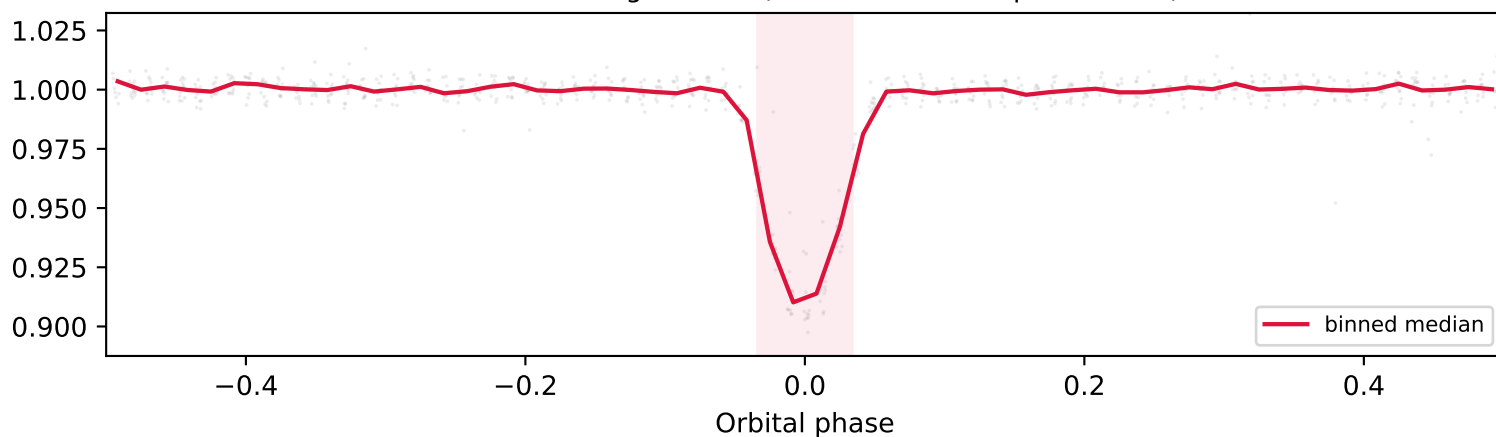
Eclipses observed 11 (gate ≥ 3)
Robust depth ratio 0.98 (gate ≥ 0.50)
Robust depth sigma 114.9 (gate ≥ 7.0)
Odd/even z-score 0.16 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 6 light-curve dip in this sector: 133.0 sigma t0 refined by 0.0100 phase
Difference-image peak SNR 57.87 (gate ≥ 5.0) Compactness 0.91 (gate ≥ 0.50)
Dimming centroid offset 0.11 px (2 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 281310326
2.04 px away, dTmag +1.77 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.

TIC 318887646

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS		CATALOG CROSS-MATCH (novelty)	
Period	14.281664 d	VSX	no match
Epoch (t0)	1418.6279 BTJD	ASAS-SN	no match
Duration	0.1200 d	Gaia eclipsing-binary table	no match
Depth	3.605 % (36,049 ppm)	Gaia variable classification	no match
SNR	149.9	TESS EB catalog (Prsa+2022)	no match
INDEPENDENT CHECKS (f3i gates)		ExoFOP TOI list	no match
Eclipses observed	3 (gate >= 3)	Verdict:	NOVEL
Robust depth ratio	1.20 (gate >= 0.50)		
Robust depth sigma	143.4 (gate >= 7.0)		
Odd/even z-score	1.52 (consistent)		
Secondary eclipse	not detected		

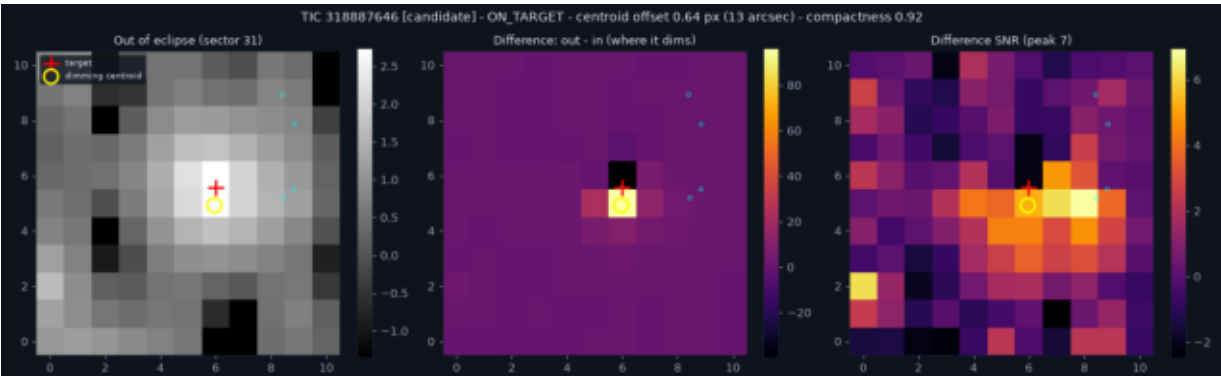
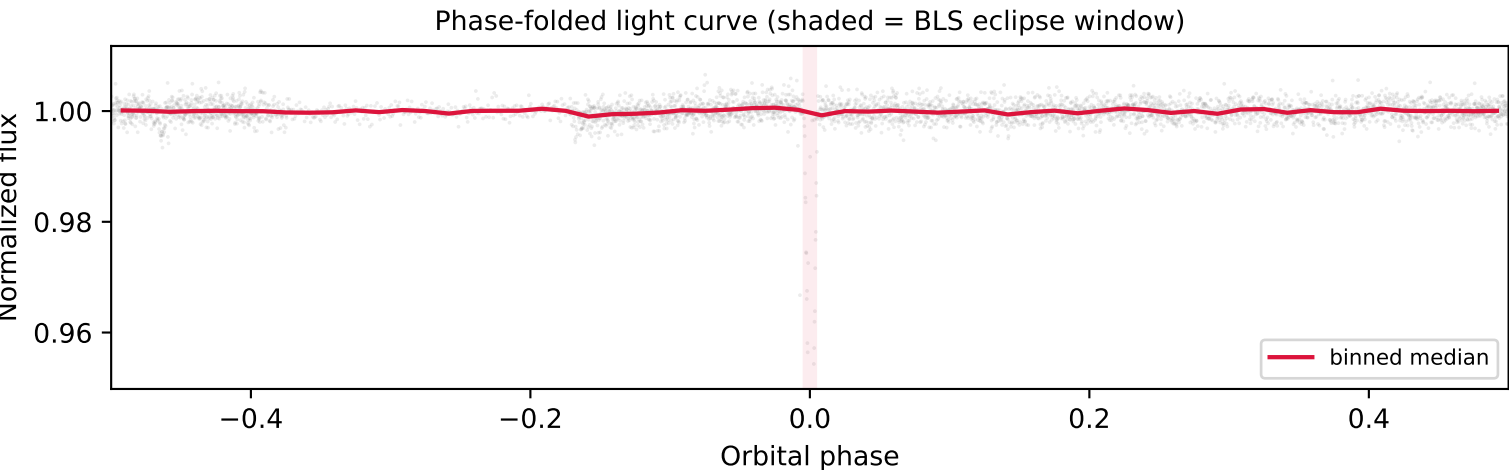
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 31 light-curve dip in this sector: 124.6 sigma t0 refined by 0.0000 phase

Difference-image peak SNR 6.94 (gate >= 5.0) Compactness 0.92 (gate >= 0.50)

Dimming centroid offset 0.64 px (13 arcsec) (gate <= 1.0 px, i.e. on the target)

Nearest bright-enough neighbour in cutout: none within the cutout



TIC 444465823

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

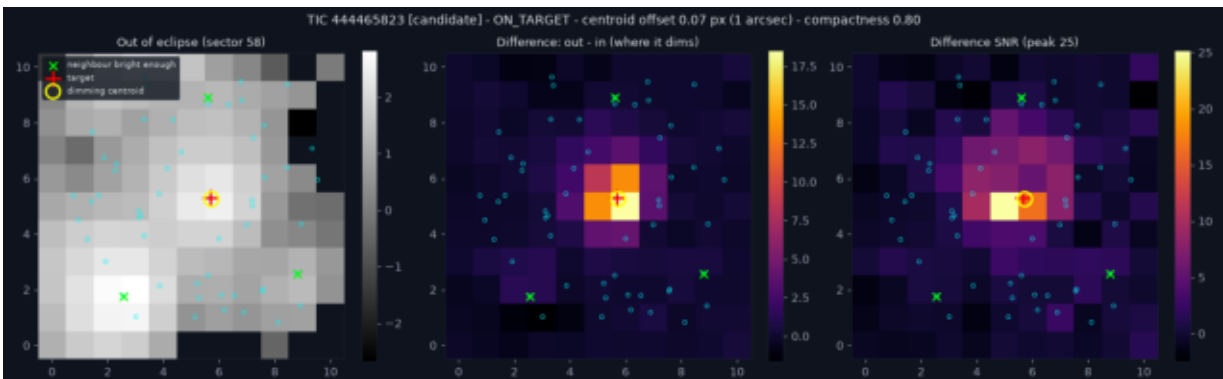
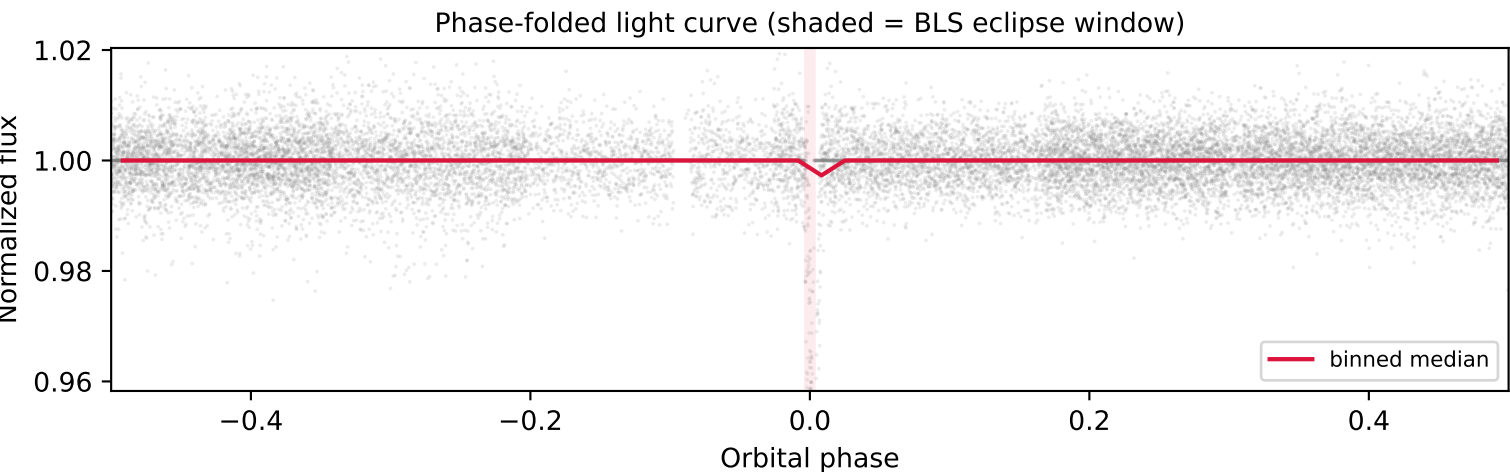
BLS EPHEMERIS
Period 18.513330 d (ALIASED, see below)
-> corrected period: 37.026660 d
Epoch (t0) 1768.5512 BTJD
Duration 0.1200 d
Depth 3.476 % (34,755 ppm)
SNR 146.6

CATALOG CROSS-MATCH (novelty)
VSX no match
ASAS-SN no match
Gaia eclipsing-binary table no match
Gaia variable classification no match
TESS EB catalog (Prsa+2022) no match
ExoFOP TOI list no match

INDEPENDENT CHECKS (f3i gates)
Eclipses observed 8 (gate >= 3)
Robust depth ratio 1.01 (gate >= 0.50)
Robust depth sigma 95.7 (gate >= 7.0)
Odd/even z-score 14.96 (ALIAS FLAGGED)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
Sector 58 light-curve dip in this sector: 61.8 sigma t0 refined by 0.0025 phase
Difference-image peak SNR 25.18 (gate >= 5.0) Compactness 0.80 (gate >= 0.50)
Dimming centroid offset 0.07 px (1 arcsec) (gate <= 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 444465799
3.65 px away, dTmag +2.93 - farther from the dimming than the target



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 14.96 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 37.026660 d, in any write-up.

TIC 380943251

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 0.720935 d
Epoch (t0) 1386.4490 BTJD
Duration 0.0580 d
Depth 7.667 % (76,670 ppm)
SNR 101.0

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

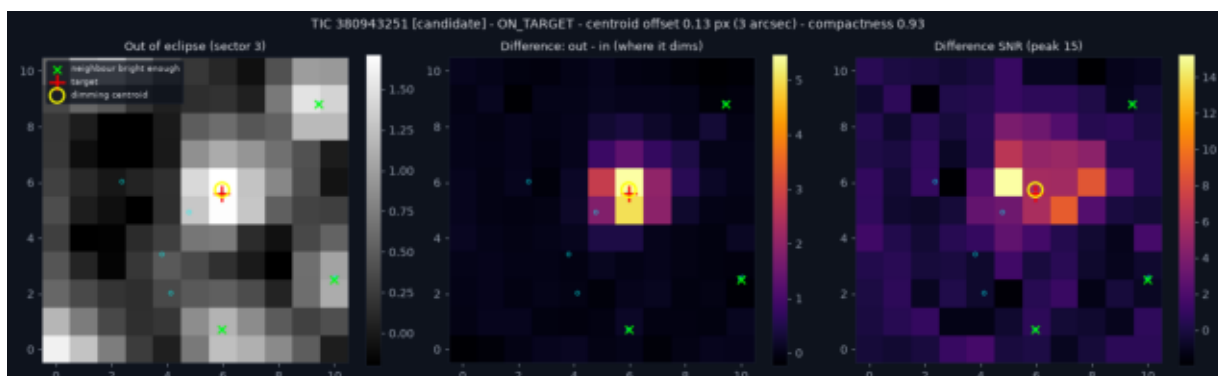
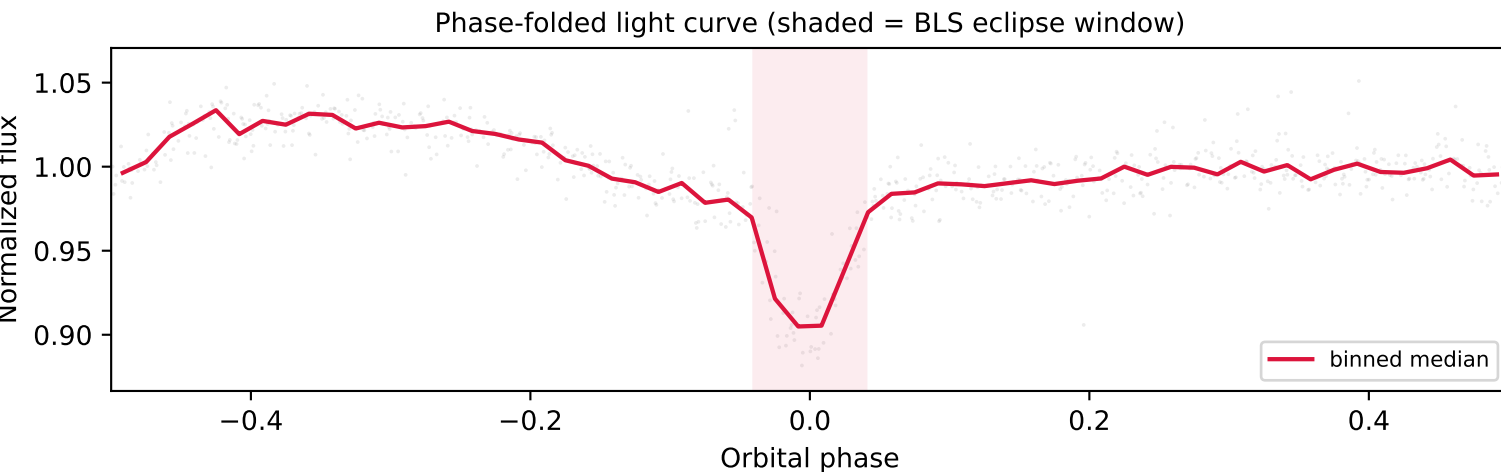
INDEPENDENT CHECKS (f3i gates)

Eclipses observed 14 (gate ≥ 3)
Robust depth ratio 1.03 (gate ≥ 0.50)
Robust depth sigma 27.8 (gate ≥ 7.0)
Odd/even z-score 1.89 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 3 light-curve dip in this sector: 25.9 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 15.26 (gate ≥ 5.0) Compactness 0.93 (gate ≥ 0.50)
Dimming centroid offset 0.13 px (3 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 380941073
4.65 px away, dTmag +0.59 - farther from the dimming than the target



TIC 66716482

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 8.028259 d
Epoch (t0) 2561.2964 BTJD
Duration 0.1200 d
Depth 3.329 % (33,287 ppm)
SNR 64.9

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

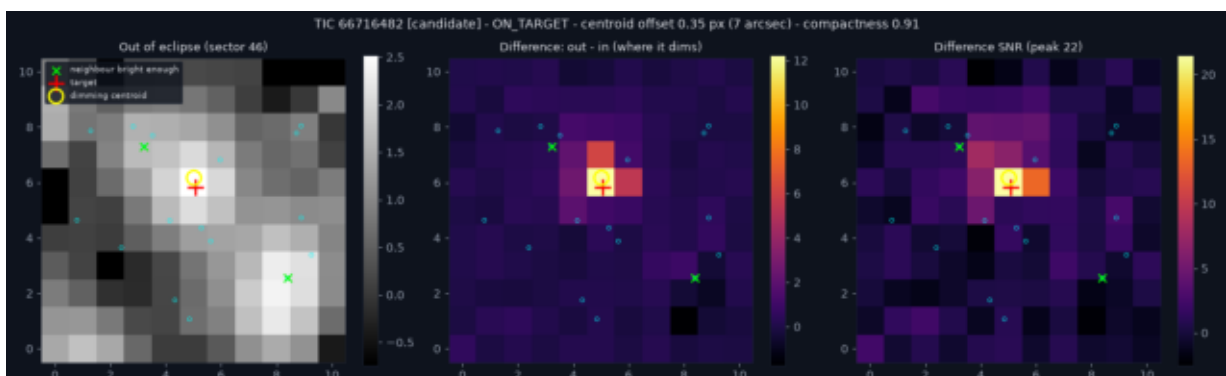
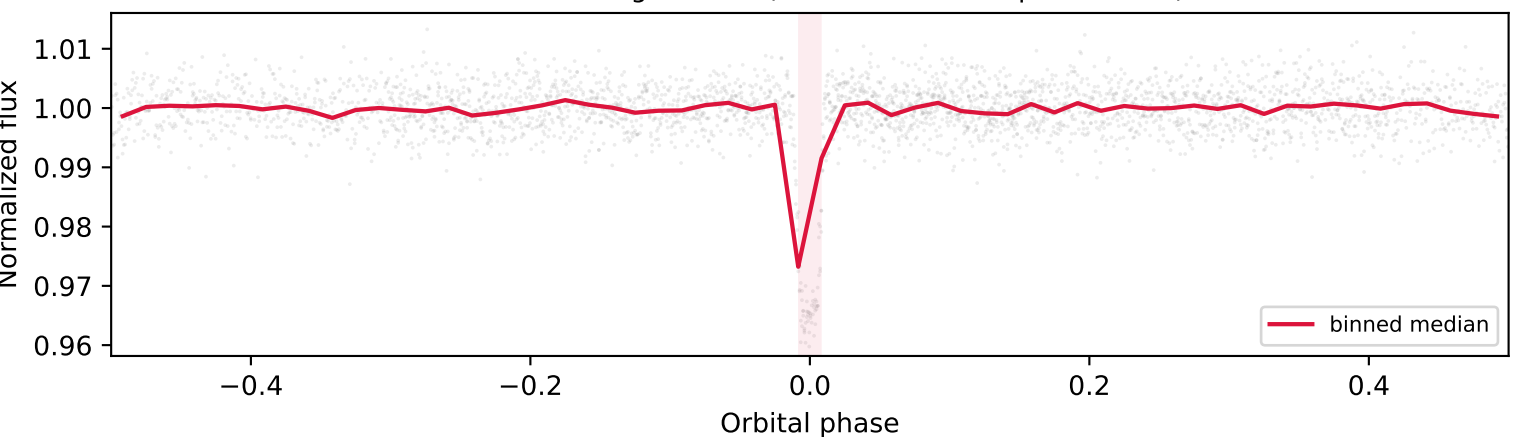
Eclipses observed 3 (gate ≥ 3)
Robust depth ratio 1.03 (gate ≥ 0.50)
Robust depth sigma 56.0 (gate ≥ 7.0)
Odd/even z-score 0.64 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 46 light-curve dip in this sector: 44.5 sigma t0 refined by -0.0025 phase
Difference-image peak SNR 21.51 (gate ≥ 5.0) Compactness 0.91 (gate ≥ 0.50)
Dimming centroid offset 0.35 px (7 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 66715933
2.13 px away, dTmag +2.34 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.

TIC 309656476

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 28.920445 d
Epoch (t0) 2997.6132 BTJD
Duration 0.1700 d
Depth 0.050 % (495 ppm)
SNR 24.6

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

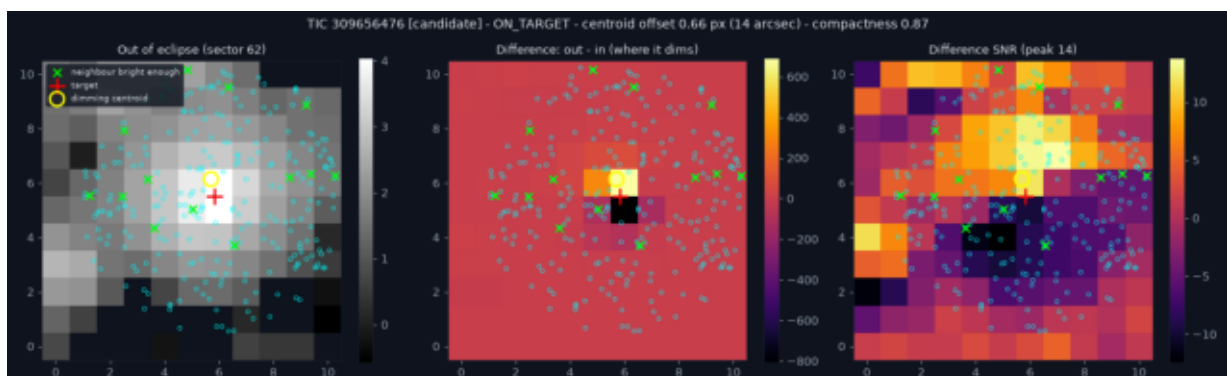
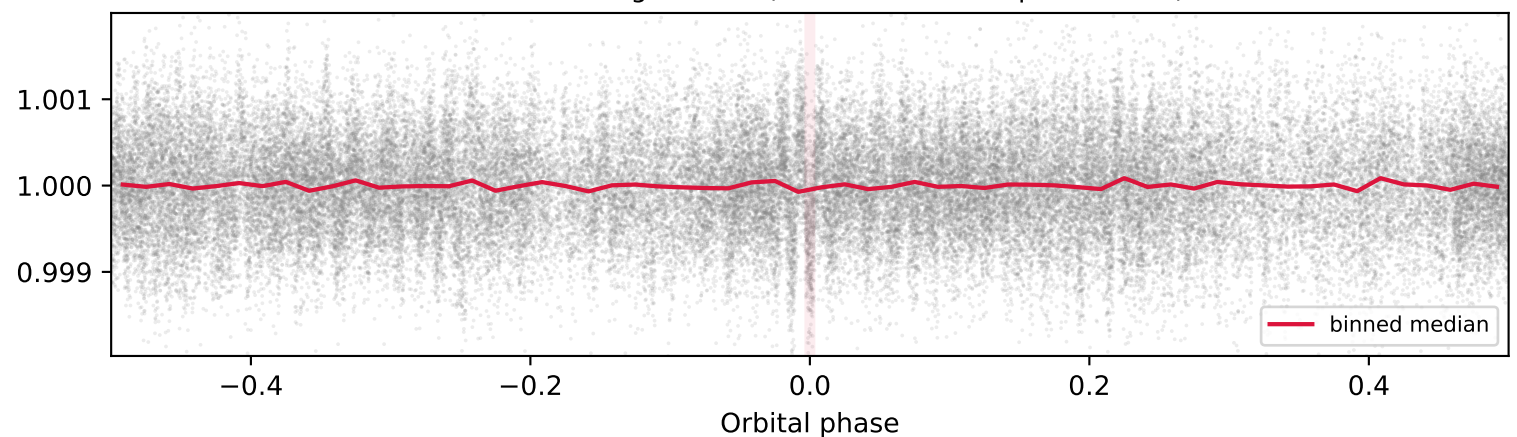
Eclipses observed 6 (gate ≥ 3)
Robust depth ratio 0.98 (gate ≥ 0.50)
Robust depth sigma 14.1 (gate ≥ 7.0)
Odd/even z-score 1.97 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 62 light-curve dip in this sector: 8.5 sigma t0 refined by 0.3975 phase
Difference-image peak SNR 13.80 (gate ≥ 5.0) Compactness 0.87 (gate ≥ 0.50)
Dimming centroid offset 0.66 px (14 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 309656481
1.29 px away, dTmag +5.82 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.

TIC 229494833

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 15.383868 d (ALIASED, see below)
-> corrected period: 30.767736 d
Epoch (t0) 2829.9134 BTJD
Duration 0.1200 d
Depth 0.037 % (372 ppm)
SNR 14.9

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

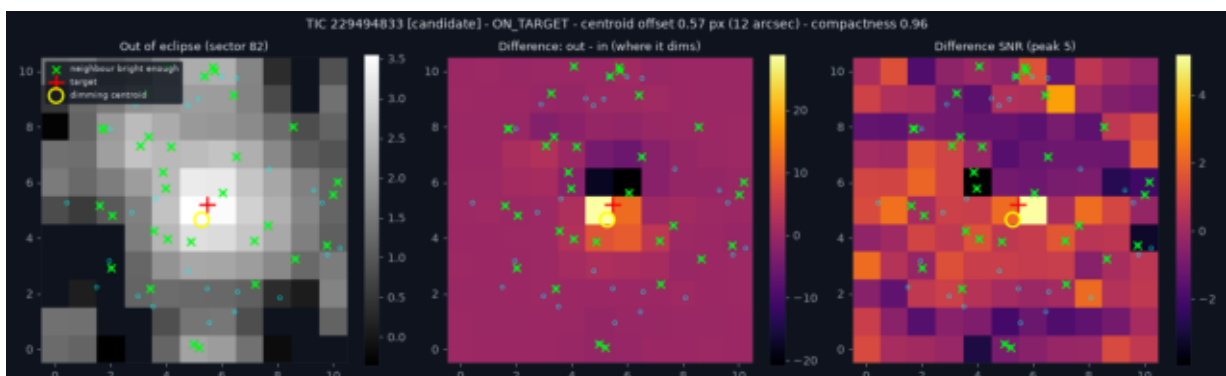
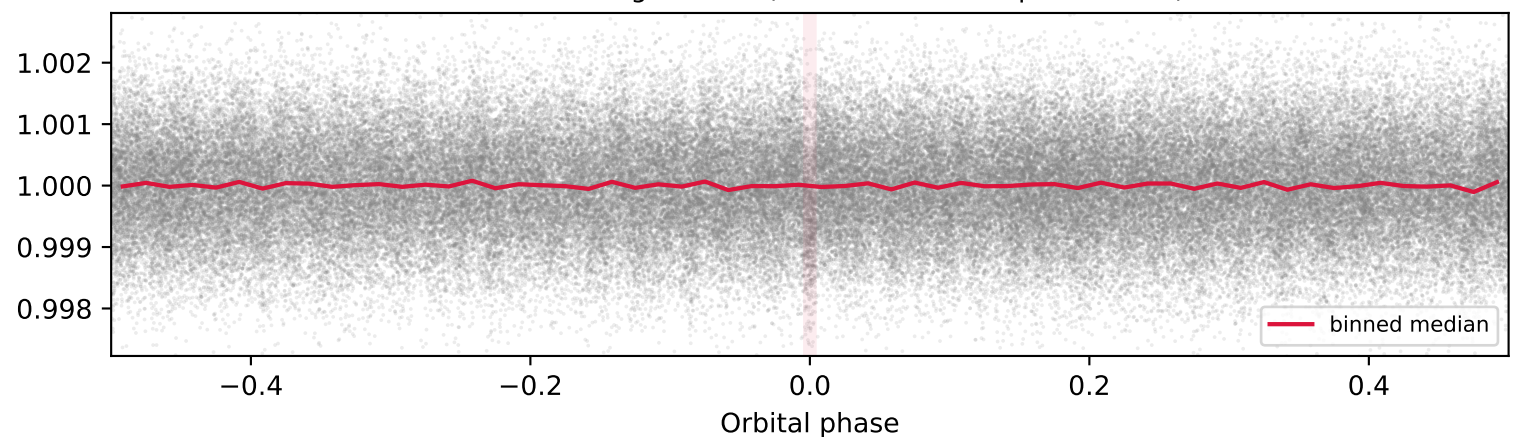
Eclipses observed 19 (gate ≥ 3)
Robust depth ratio 1.00 (gate ≥ 0.50)
Robust depth sigma 11.2 (gate ≥ 7.0)
Odd/even z-score 4.32 (ALIAS FLAGGED)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 82 light-curve dip in this sector: 6.8 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 5.19 (gate ≥ 5.0) Compactness 0.96 (gate ≥ 0.50)
Dimming centroid offset 0.57 px (12 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 229494826
0.88 px away, dTmag +4.84 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 4.32 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 30.767736 d, in any write-up.

TIC 44533527

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 14.595189 d
Epoch (t0) 3463.8871 BTJD
Duration 0.1200 d
Depth 0.101 % (1,013 ppm)
SNR 12.0

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

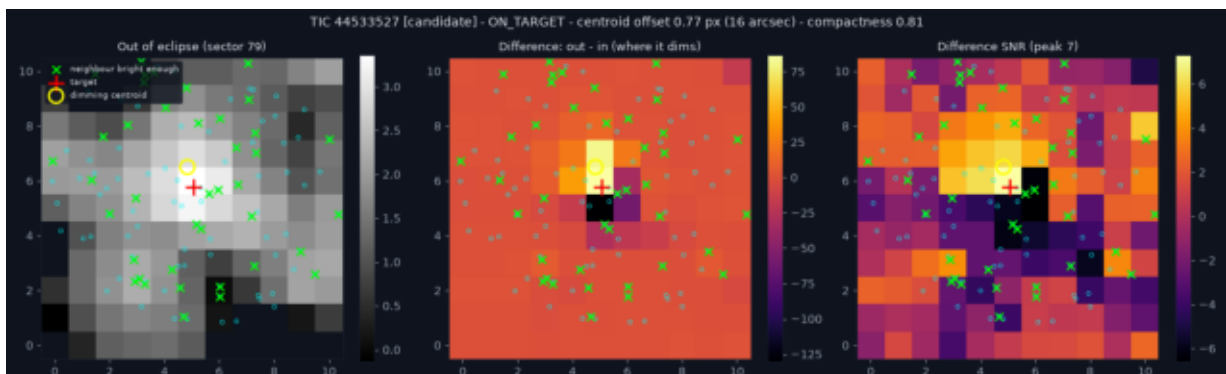
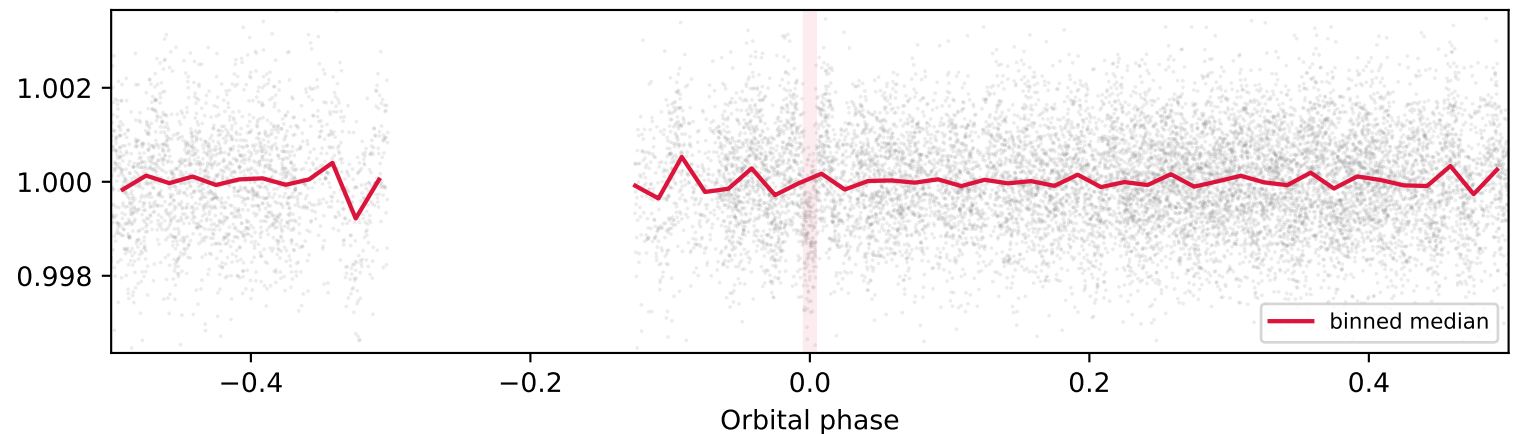
Eclipses observed 3 (gate ≥ 3)
Robust depth ratio 0.94 (gate ≥ 0.50)
Robust depth sigma 8.6 (gate ≥ 7.0)
Odd/even z-score 1.18 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 79 light-curve dip in this sector: 6.8 sigma t0 refined by -0.3250 phase
Difference-image peak SNR 7.33 (gate ≥ 5.0) Compactness 0.81 (gate ≥ 0.50)
Dimming centroid offset 0.77 px (16 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 44533532
1.26 px away, dTmag +4.17 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



TIC 220423668

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 3.651055 d (ALIASED, see below)
-> corrected period: 7.302111 d
Epoch (t0) 1413.9498 BTJD
Duration 0.1200 d
Depth 4.525 % (45,247 ppm)
SNR 755.4

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

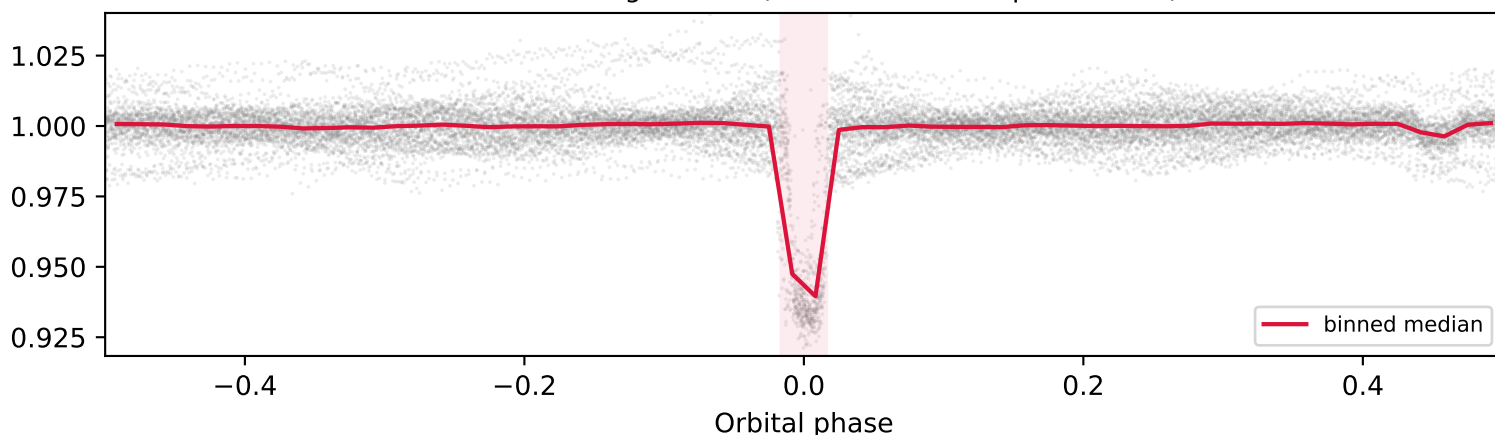
Eclipses observed 55 (gate ≥ 3)
Robust depth ratio 1.27 (gate ≥ 0.50)
Robust depth sigma 215.5 (gate ≥ 7.0)
Odd/even z-score 8.94 (ALIAS FLAGGED)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 32 light-curve dip in this sector: 83.6 sigma t0 refined by 0.0025 phase
Difference-image peak SNR 17.67 (gate ≥ 5.0) Compactness 0.75 (gate ≥ 0.50)
Dimming centroid offset 0.28 px (6 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 220423674
3.23 px away, dTmag +1.58 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



difference-image figure not found

TIC 365998904

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 1.249358 d
Epoch (t0) 1654.4709 BTJD
Duration 0.1200 d
Depth 2.398 % (23,984 ppm)
SNR 375.0

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

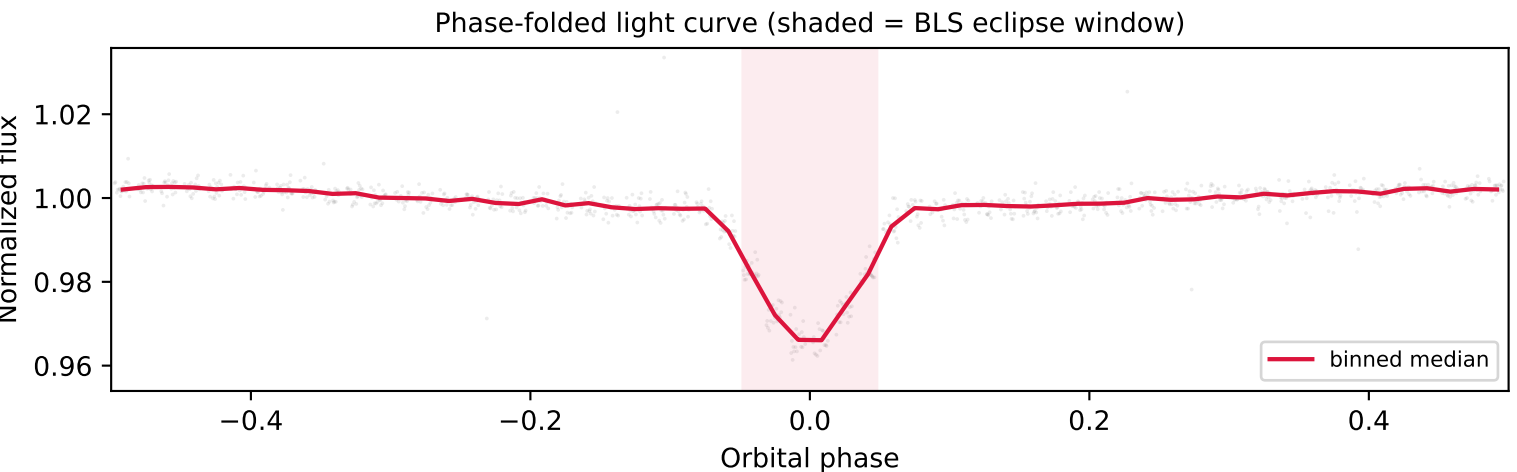
INDEPENDENT CHECKS (f3i gates)

Eclipses observed 22 (gate ≥ 3)
Robust depth ratio 1.12 (gate ≥ 0.50)
Robust depth sigma 87.4 (gate ≥ 7.0)
Odd/even z-score 0.27 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 13 light-curve dip in this sector: 108.5 sigma t0 refined by -0.0050 phase
Difference-image peak SNR 15.52 (gate ≥ 5.0) Compactness 0.86 (gate ≥ 0.50)
Dimming centroid offset 0.14 px (3 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 365998896
2.59 px away, dTmag -1.63 - farther from the dimming than the target



difference-image figure not found

BLS EPHEMERIS		CATALOG CROSS-MATCH (novelty)	
Period	1.906321 d	VSX	no match
Epoch (t0)	1625.2491 BTJD	ASAS-SN	no match
Duration	0.3500 d	Gaia eclipsing-binary table	no match
Depth	5.467 % (54,673 ppm)	Gaia variable classification	no match
SNR	202.5	TESS EB catalog (Prsa+2022)	no match
INDEPENDENT CHECKS (f3i gates)		ExoFOP TOI list	no match
Eclipses observed	14 (gate >= 3)	Verdict: NOVEL	
Robust depth ratio	1.08 (gate >= 0.50)		
Robust depth sigma	56.5 (gate >= 7.0)		
Odd/even z-score	0.02 (consistent)		
Secondary eclipse	not detected		

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

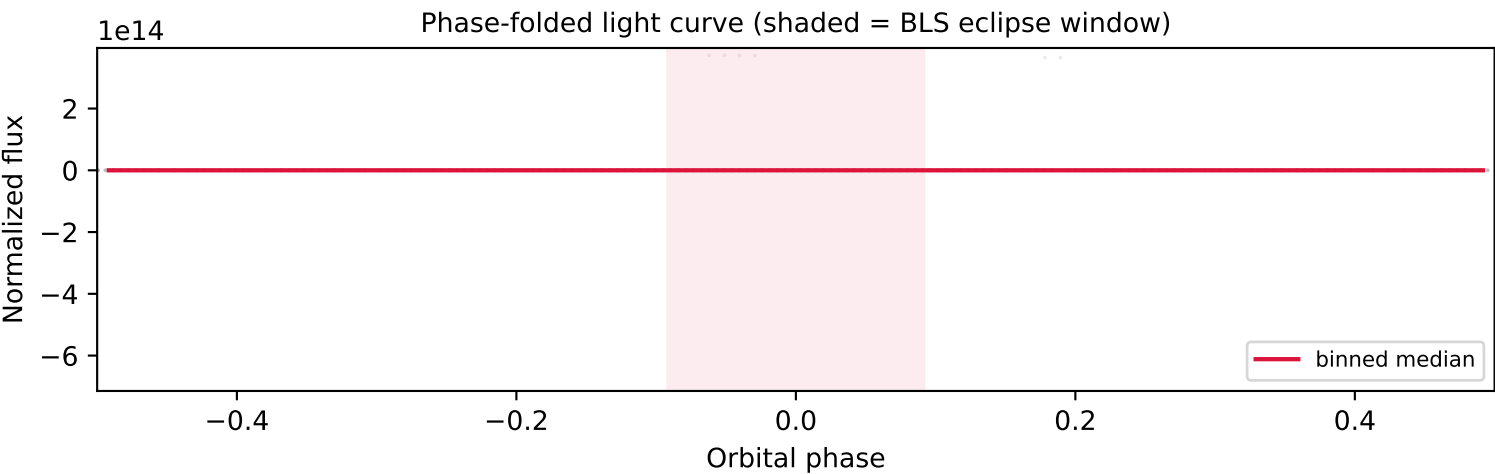
Sector 12 light-curve dip in this sector: 57.1 sigma t0 refined by 0.0050 phase

Difference-image peak SNR 8.94 (gate >= 5.0) Compactness 1.00 (gate >= 0.50)

Dimming centroid offset 0.08 px (2 arcsec) (gate <= 1.0 px, i.e. on the target)

Nearest bright-enough neighbour in cutout: TIC 1332737174

0.11 px away, dTmag +3.07 - farther from the dimming than the target



difference-image figure not found

Caveat (THIN MARGIN): The ON_TARGET call rests on a neighbour-competition margin of just 0.030 px (dimming centroid 0.083 px from target vs 0.113 px from the nearest bright-enough neighbour) - inside diag_pixel.py's own stated centroid systematic of 0.10-0.3 px. TESS's resolution cannot confidently separate this star from that neighbour.

TIC 437171483

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 1.473514 d
Epoch (t0) 1543.8678 BTJD
Duration 0.0840 d
Depth 4.947 % (49,472 ppm)
SNR 922.8

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

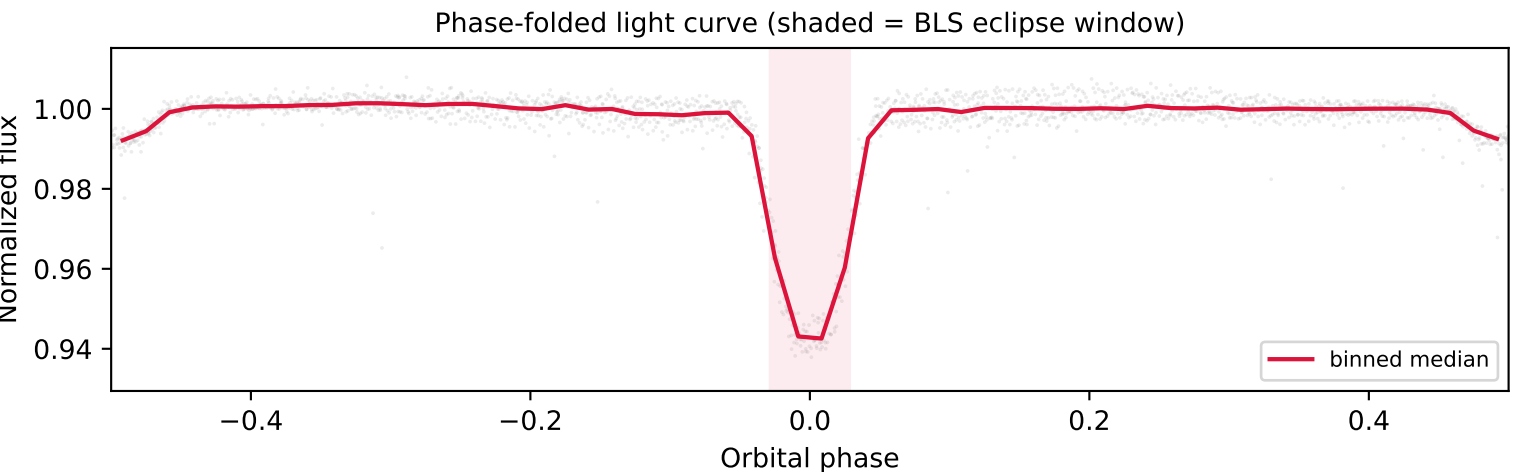
INDEPENDENT CHECKS (f3i gates)

Eclipses observed 30 (gate ≥ 3)
Robust depth ratio 1.06 (gate ≥ 0.50)
Robust depth sigma 173.5 (gate ≥ 7.0)
Odd/even z-score 1.65 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 9 light-curve dip in this sector: 127.7 sigma t0 refined by -0.0025 phase
Difference-image peak SNR 73.37 (gate ≥ 5.0) Compactness 0.91 (gate ≥ 0.50)
Dimming centroid offset 0.14 px (3 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 437171484
0.55 px away, dTmag +3.05 - farther from the dimming than the target



difference-image figure not found

TIC 411457490

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 2.432352 d (ALIASSED, see below)
-> corrected period: 4.864705 d
Epoch (t0) 1596.8966 BTJD
Duration 0.1700 d
Depth 5.531 % (55,313 ppm)
SNR 431.2

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

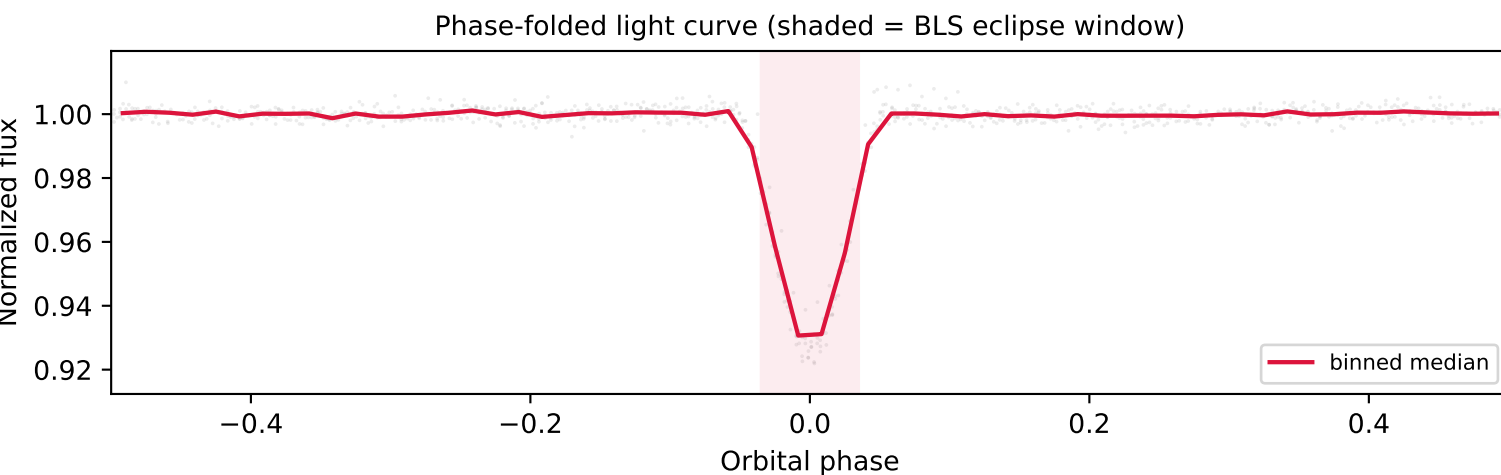
INDEPENDENT CHECKS (f3i gates)

Eclipses observed 12 (gate ≥ 3)
Robust depth ratio 1.06 (gate ≥ 0.50)
Robust depth sigma 243.6 (gate ≥ 7.0)
Odd/even z-score 4.70 (ALIAS FLAGGED)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 11 light-curve dip in this sector: 231.3 sigma t0 refined by 0.0100 phase
Difference-image peak SNR 58.00 (gate ≥ 5.0) Compactness 0.90 (gate ≥ 0.50)
Dimming centroid offset 0.03 px (1 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 411457450
0.94 px away, dTmag +2.68 - farther from the dimming than the target



difference-image figure not found

TIC 295373509

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 5.421872 d
Epoch (t0) 1654.9396 BTJD
Duration 0.1700 d
Depth 8.026 % (80,259 ppm)
SNR 246.8

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

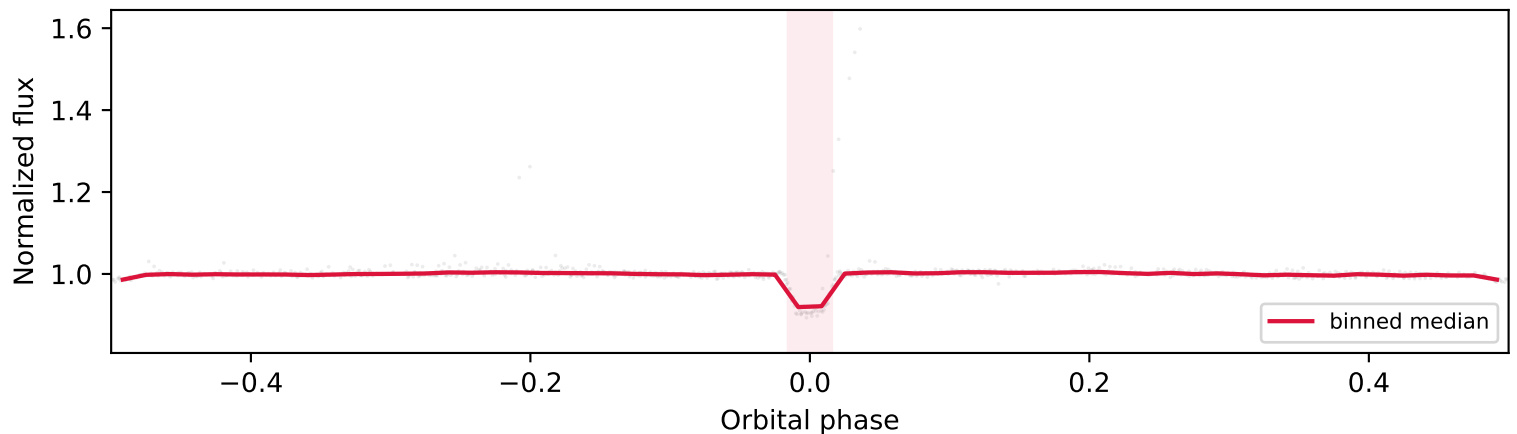
Eclipses observed 6 (gate ≥ 3)
Robust depth ratio 1.01 (gate ≥ 0.50)
Robust depth sigma 86.4 (gate ≥ 7.0)
Odd/even z-score 0.02 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 13 light-curve dip in this sector: 83.4 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 22.12 (gate ≥ 5.0) Compactness 0.87 (gate ≥ 0.50)
Dimming centroid offset 0.07 px (2 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 295373523
3.19 px away, dTmag +2.13 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



difference-image figure not found

TIC 374158907

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 14.656290 d (ALIASED, see below)
-> corrected period: 29.312581 d
Epoch (t0) 1551.5656 BTJD
Duration 0.1200 d
Depth 4.974 % (49,740 ppm)
SNR 178.6

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

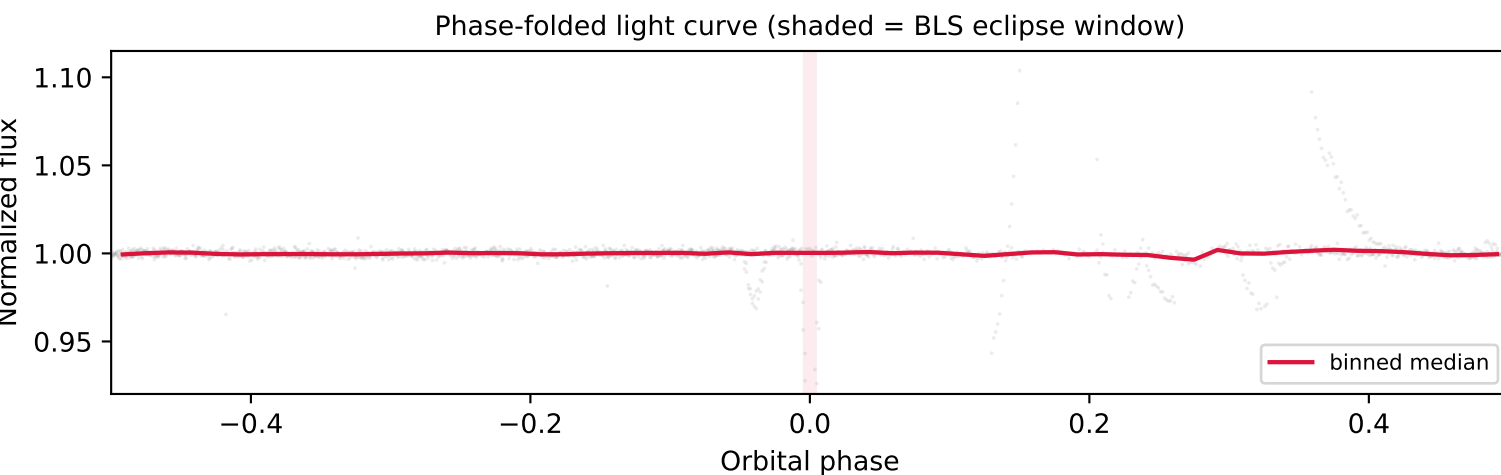
INDEPENDENT CHECKS (f3i gates)

Eclipses observed 4 (gate ≥ 3)
Robust depth ratio 1.14 (gate ≥ 0.50)
Robust depth sigma 125.0 (gate ≥ 7.0)
Odd/even z-score 14.90 (ALIAS FLAGGED)
Secondary eclipse detected, phase 0.26, 7.4 sigma

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 9 light-curve dip in this sector: 82.6 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 21.40 (gate ≥ 5.0) Compactness 0.77 (gate ≥ 0.50)
Dimming centroid offset 0.10 px (2 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 374158897
1.05 px away, dTmag -0.74 - farther from the dimming than the target



difference-image figure not found

TIC 348884775

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 6.731631 d
Epoch (t0) 1519.7050 BTJD
Duration 0.1700 d
Depth 14.751 % (147,512 ppm)
SNR 111.3

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

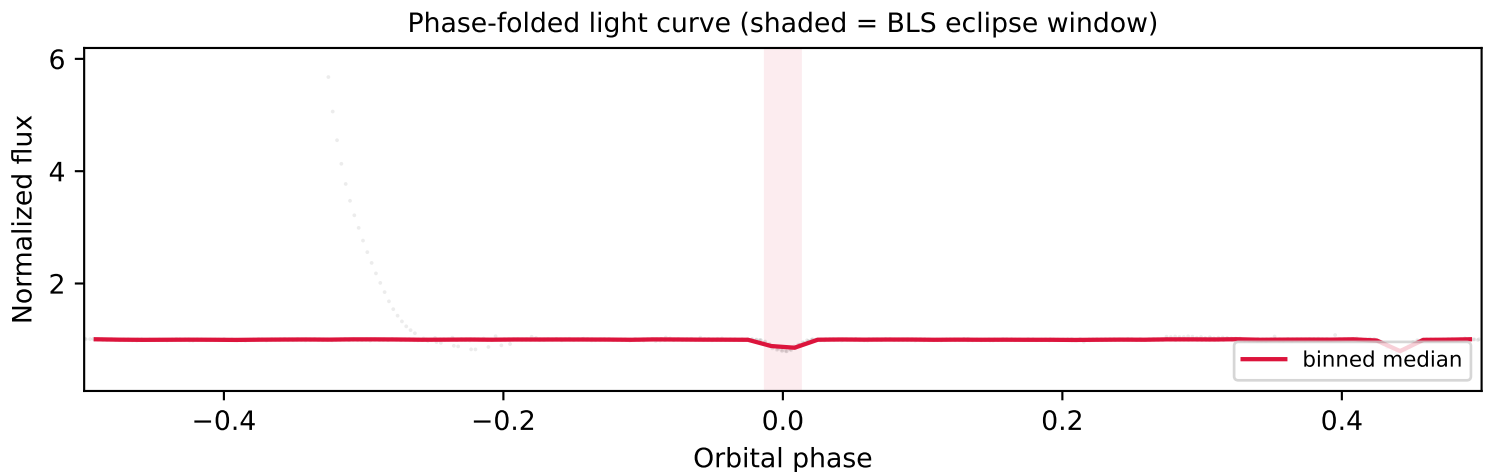
INDEPENDENT CHECKS (f3i gates)

Eclipses observed 3 (gate ≥ 3)
Robust depth ratio 1.04 (gate ≥ 0.50)
Robust depth sigma 80.0 (gate ≥ 7.0)
Odd/even z-score 0.39 (consistent)
Secondary eclipse detected, phase 0.43, 7.3 sigma

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 8 light-curve dip in this sector: 66.5 sigma t0 refined by 0.4425 phase
Difference-image peak SNR 17.82 (gate ≥ 5.0) Compactness 0.71 (gate ≥ 0.50)
Dimming centroid offset 0.20 px (4 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 348884765
4.70 px away, dTmag +0.97 - farther from the dimming than the target



difference-image figure not found

TIC 70762350

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 3.791085 d
Epoch (t0) 1355.3548 BTJD
Duration 0.1200 d
Depth 7.147 % (71,472 ppm)
SNR 66.2

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

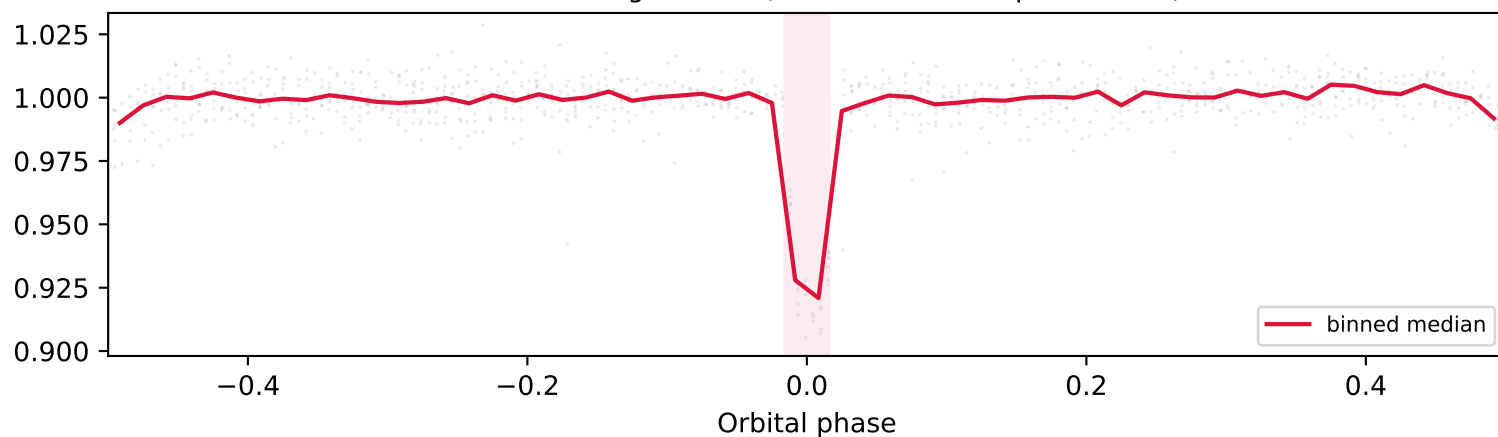
Eclipses observed 7 (gate ≥ 3)
Robust depth ratio 1.04 (gate ≥ 0.50)
Robust depth sigma 51.1 (gate ≥ 7.0)
Odd/even z-score 0.70 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 2 light-curve dip in this sector: 43.1 sigma t0 refined by 0.0025 phase
Difference-image peak SNR 15.20 (gate ≥ 5.0) Compactness 0.93 (gate ≥ 0.50)
Dimming centroid offset 0.15 px (3 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 70762352
2.01 px away, dTmag +2.45 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



difference-image figure not found

BLS EPHEMERIS		CATALOG CROSS-MATCH (novelty)	
Period	1.107957 d	VSX	no match
Epoch (t0)	3480.6522 BTJD	ASAS-SN	no match
Duration	0.1200 d	Gaia eclipsing-binary table	no match
Depth	0.231 % (2,310 ppm)	Gaia variable classification	no match
SNR	12.5	TESS EB catalog (Prsa+2022)	no match
INDEPENDENT CHECKS (f3i gates)		ExoFOP TOI list	no match
Eclipses observed	16 (gate >= 3)	Verdict: NOVEL	
Robust depth ratio	0.91 (gate >= 0.50)		
Robust depth sigma	9.9 (gate >= 7.0)		
Odd/even z-score	0.18 (consistent)		
Secondary eclipse	not detected		

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

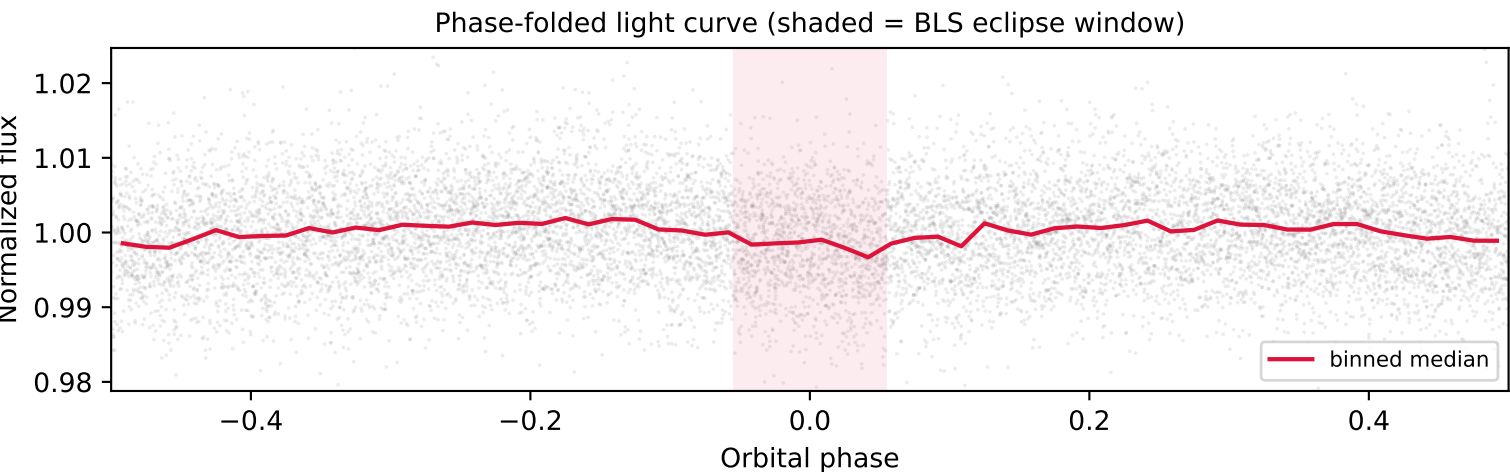
Sector 80 light-curve dip in this sector: 11.1 sigma t0 refined by 0.0300 phase

Difference-image peak SNR 6.06 (gate >= 5.0) Compactness 0.56 (gate >= 0.50)

Dimming centroid offset 0.16 px (3 arcsec) (gate <= 1.0 px, i.e. on the target)

Nearest bright-enough neighbour in cutout: TIC 1748977177

0.25 px away, dTmag +6.56 - farther from the dimming than the target



difference-image figure not found

Caveat (THIN MARGIN): The ON_TARGET call rests on a neighbour-competition margin of just 0.090 px (dimming centroid 0.158 px from target vs 0.247 px from the nearest bright-enough neighbour) - inside diag_pixel.py's own stated centroid systematic of 0.10-0.3 px. TESS's resolution cannot confidently separate this star from that neighbour.

TIC 326225355

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 15.291303 d (ALIASED, see below)
-> corrected period: 30.582605 d
Epoch (t0) 1632.8660 BTJD
Duration 0.2440 d
Depth 7.762 % (77,617 ppm)
SNR 1242.8

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

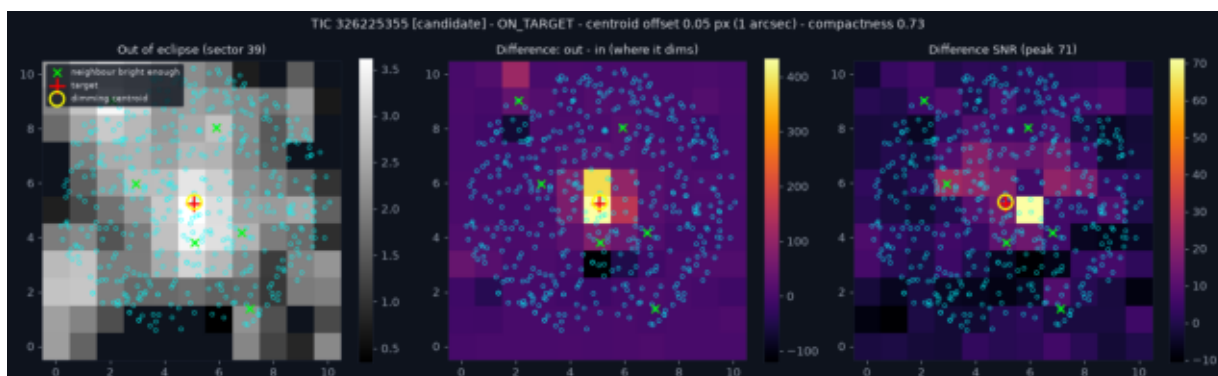
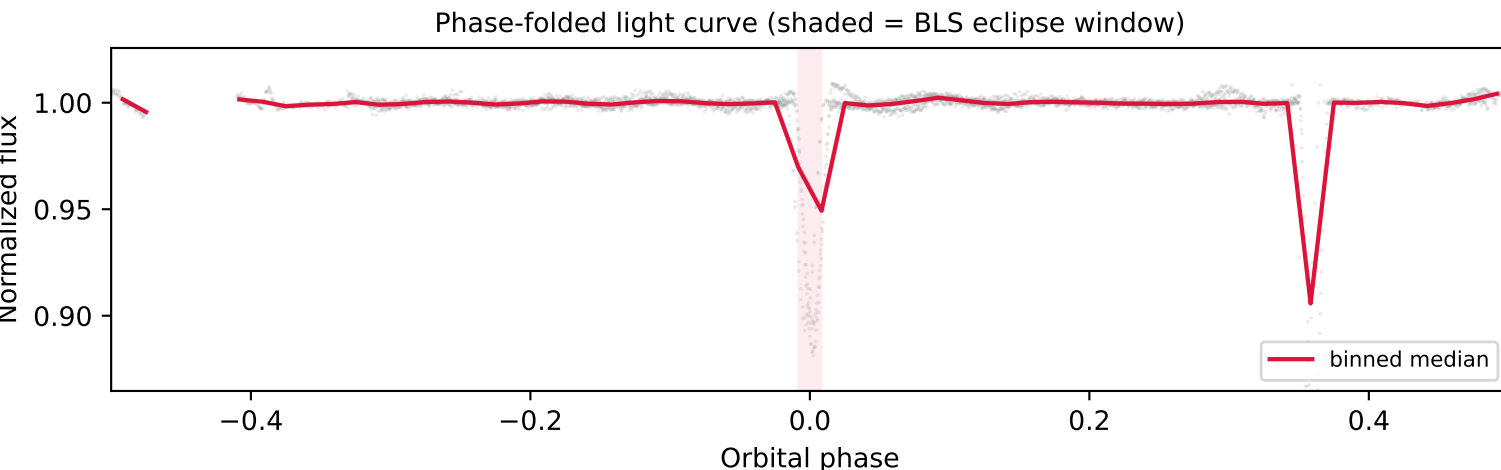
INDEPENDENT CHECKS (f3i gates)

Eclipses observed 4 (gate ≥ 3)
Robust depth ratio 1.12 (gate ≥ 0.50)
Robust depth sigma 424.9 (gate ≥ 7.0)
Odd/even z-score 3.56 (ALIAS FLAGGED)
Secondary eclipse detected, phase 0.36, 6.8 sigma

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 39 light-curve dip in this sector: 285.8 sigma t0 refined by 0.3600 phase
Difference-image peak SNR 71.43 (gate ≥ 5.0) Compactness 0.73 (gate ≥ 0.50)
Dimming centroid offset 0.05 px (1 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 326225457
1.50 px away, dTmag +0.82 - farther from the dimming than the target



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 3.56 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 30.582605 d, in any write-up.

TIC 250549396

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 2.192896 d
Epoch (t0) 1469.1898 BTJD
Duration 0.1700 d
Depth 6.895 % (68,951 ppm)
SNR 837.8

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

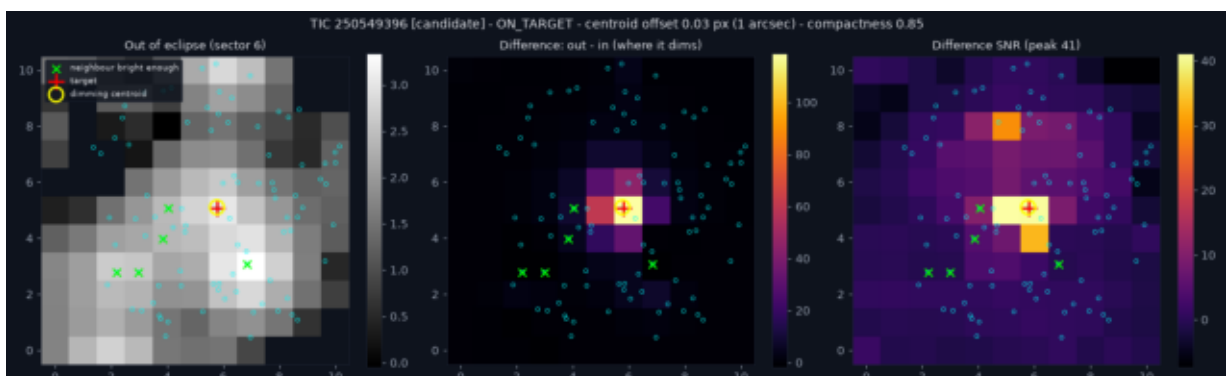
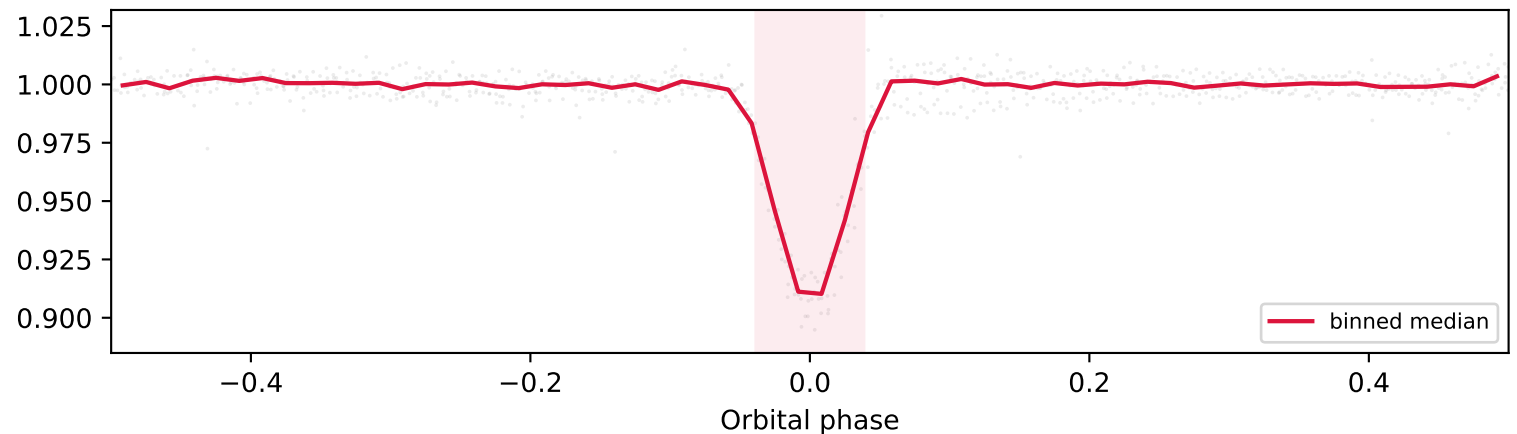
Eclipses observed 9 (gate ≥ 3)
Robust depth ratio 1.06 (gate ≥ 0.50)
Robust depth sigma 107.7 (gate ≥ 7.0)
Odd/even z-score 2.51 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 6 light-curve dip in this sector: 107.2 sigma t0 refined by 0.0025 phase
Difference-image peak SNR 41.15 (gate ≥ 5.0) Compactness 0.85 (gate ≥ 0.50)
Dimming centroid offset 0.03 px (1 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 250549395
1.73 px away, dTmag +2.30 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.

TIC 430314843

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 1.144289 d
Epoch (t0) 2474.6036 BTJD
Duration 0.0580 d
Depth 6.744 % (67,436 ppm)
SNR 638.6

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

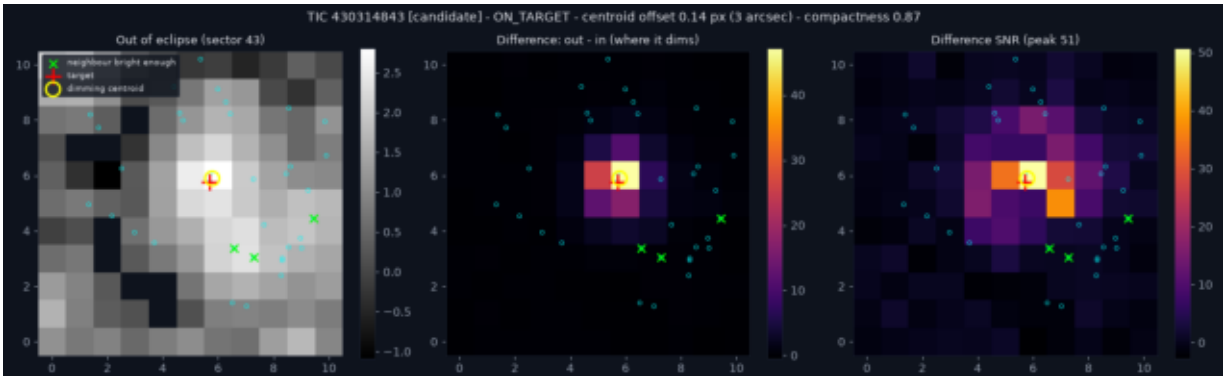
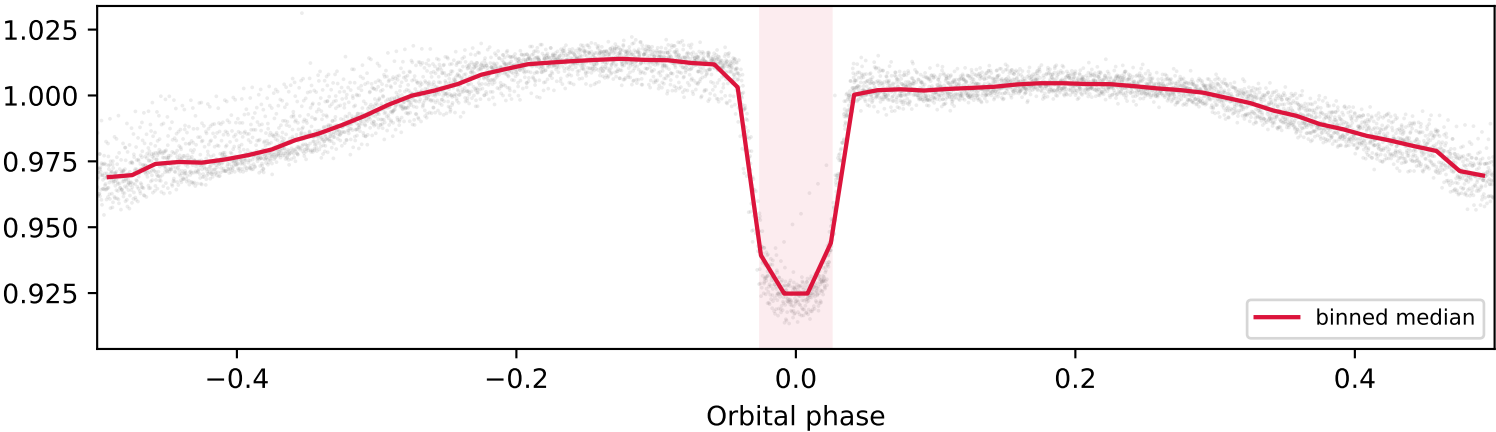
Eclipses observed 39 (gate >= 3)
Robust depth ratio 1.08 (gate >= 0.50)
Robust depth sigma 74.6 (gate >= 7.0)
Odd/even z-score 1.17 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 43 light-curve dip in this sector: 106.3 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 50.73 (gate >= 5.0) Compactness 0.87 (gate >= 0.50)
Dimming centroid offset 0.14 px (3 arcsec) (gate <= 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 397286164
2.64 px away, dTmag +1.10 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



TIC 370553309

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 11.540028 d
Epoch (t0) 1576.2791 BTJD
Duration 0.3500 d
Depth 2.621 % (26,206 ppm)
SNR 355.4

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

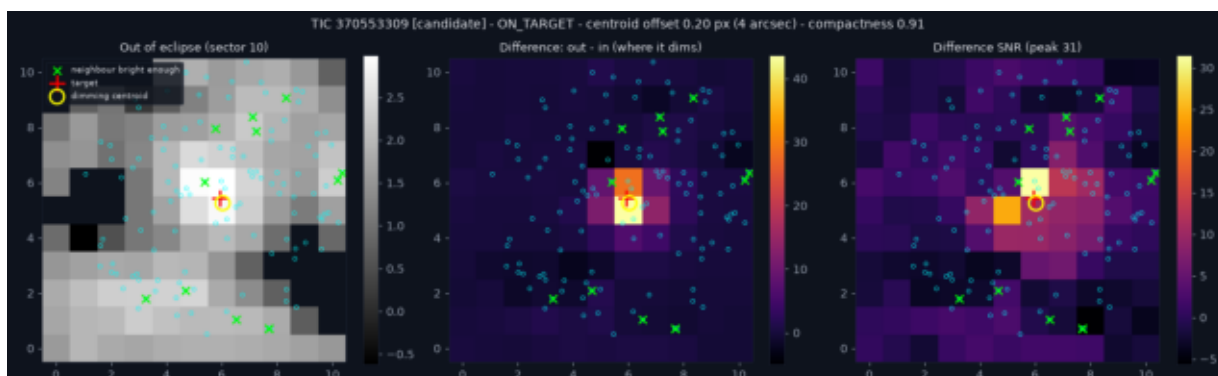
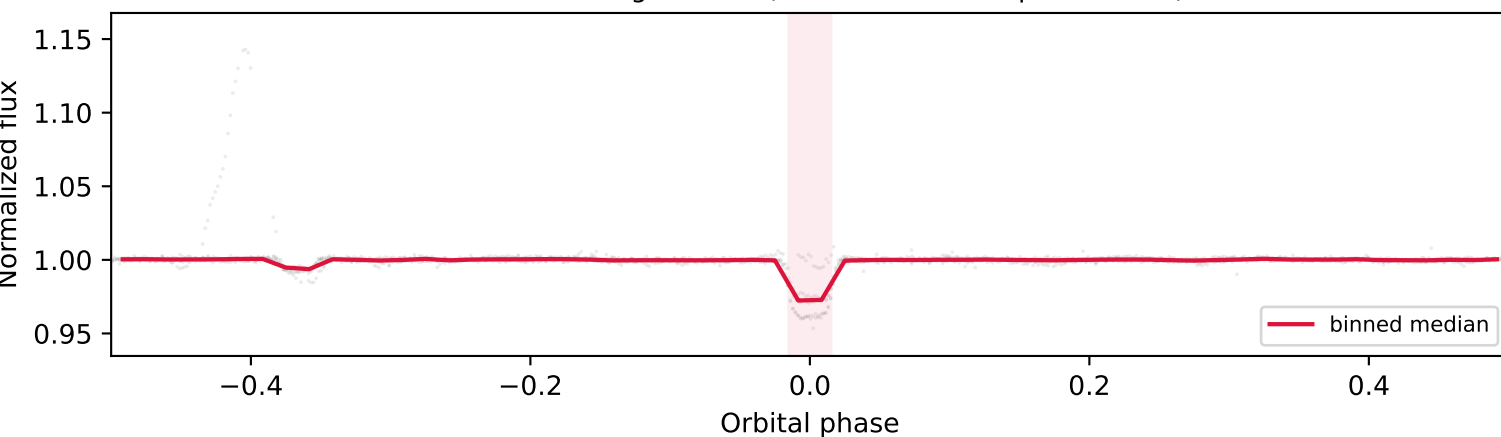
Eclipses observed 5 (gate ≥ 3)
Robust depth ratio 1.06 (gate ≥ 0.50)
Robust depth sigma 179.0 (gate ≥ 7.0)
Odd/even z-score 1.97 (consistent)
Secondary eclipse detected, phase -0.36, 5.7 sigma

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 10 light-curve dip in this sector: 126.6 sigma t0 refined by 0.0025 phase
Difference-image peak SNR 31.48 (gate ≥ 5.0) Compactness 0.91 (gate ≥ 0.50)
Dimming centroid offset 0.20 px (4 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 370553325
1.01 px away, dTmag +0.31 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



TIC 86956236

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 35.420027 d (ALIASED, see below)
-> corrected period: 70.840054 d
Epoch (t0) 1843.8893 BTJD
Duration 0.1200 d
Depth 4.974 % (49,742 ppm)
SNR 225.5

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

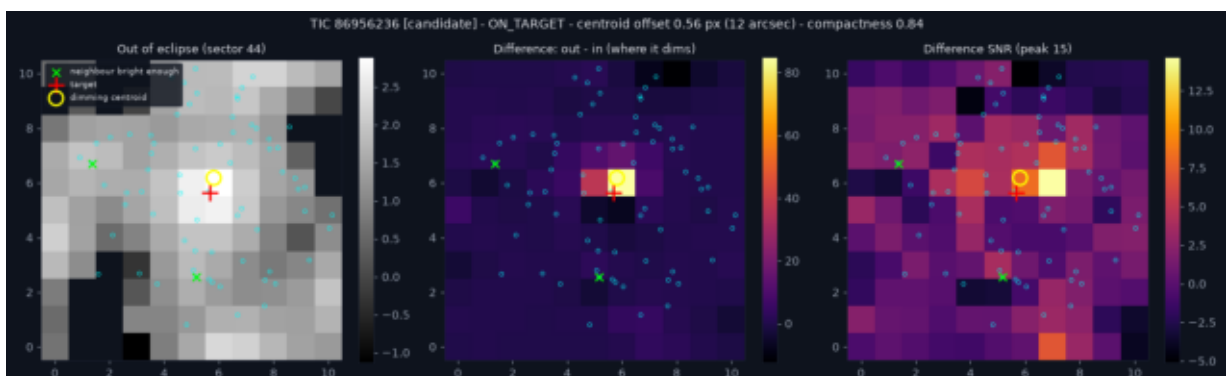
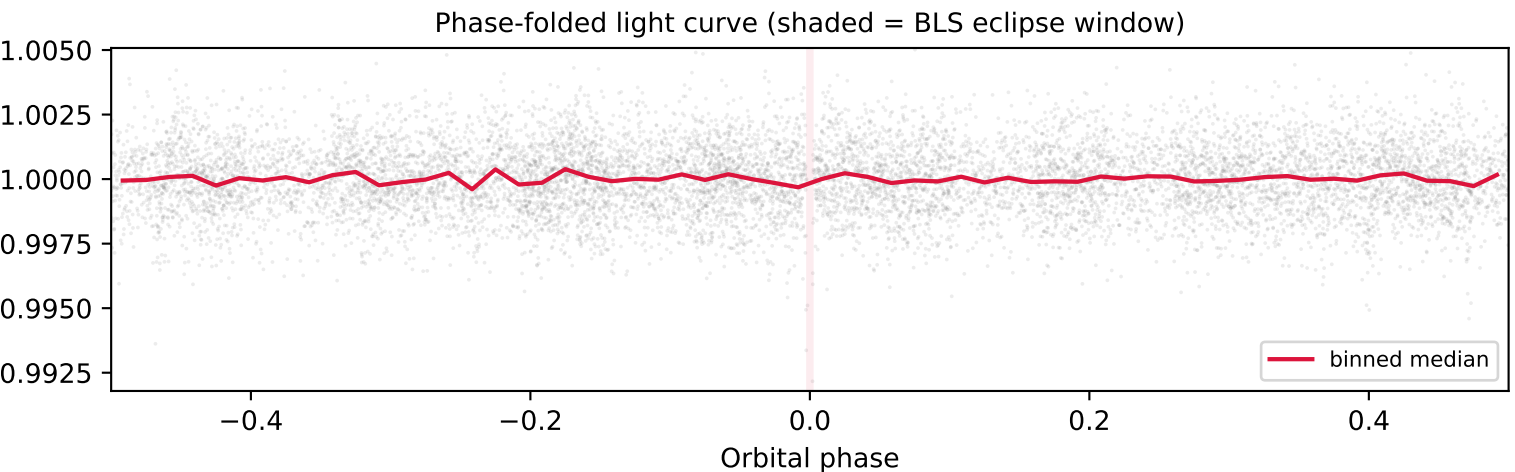
INDEPENDENT CHECKS (f3i gates)

Eclipses observed 3 (gate ≥ 3)
Robust depth ratio 1.07 (gate ≥ 0.50)
Robust depth sigma 200.3 (gate ≥ 7.0)
Odd/even z-score 6.26 (ALIAS FLAGGED)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 44 light-curve dip in this sector: 103.7 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 14.68 (gate ≥ 5.0) Compactness 0.84 (gate ≥ 0.50)
Dimming centroid offset 0.56 px (12 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 86956227
3.68 px away, dTmag +3.09 - farther from the dimming than the target



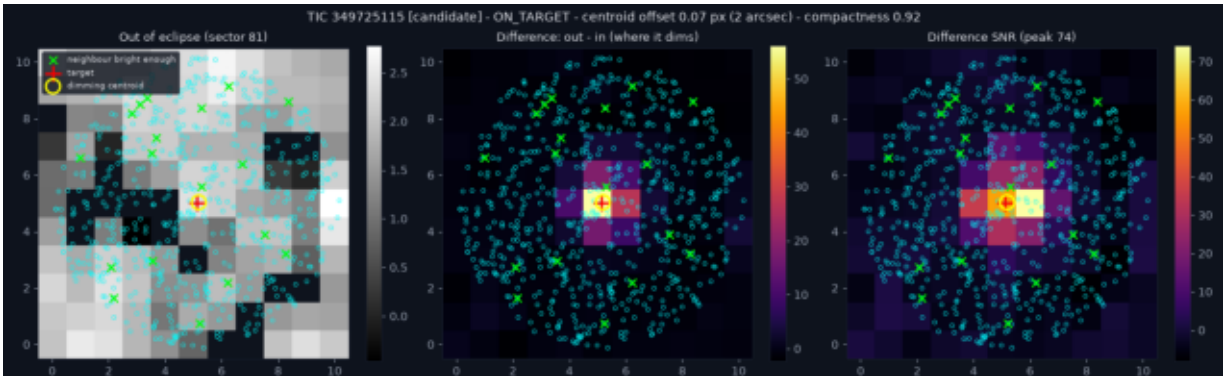
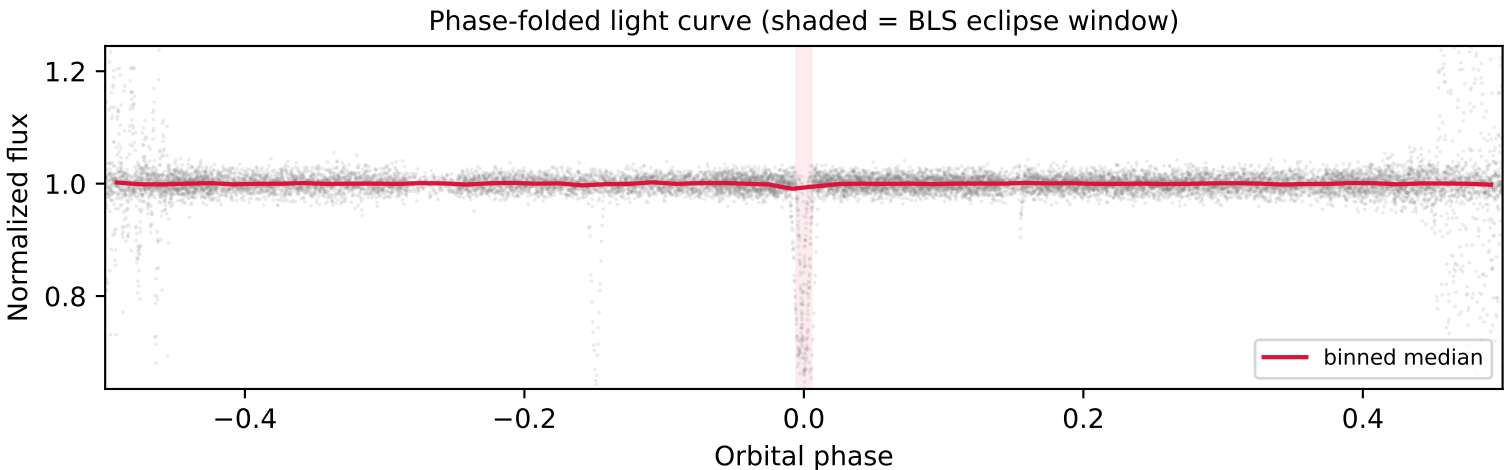
Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 6.26 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 70.840054 d, in any write-up.

TIC 349725115

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS		CATALOG CROSS-MATCH (novelty)	
Period	16.220320 d (ALIASED, see below)	VSX	no match
-> corrected period:	32.440640 d	ASAS-SN	no match
Epoch (t0)	2777.6829 BTJD	Gaia eclipsing-binary table	no match
Duration	0.1700 d	Gaia variable classification	no match
Depth	16.867 % (168,665 ppm)	TESS EB catalog (Prsa+2022)	no match
SNR	177.1	ExoFOP TOI list	no match
INDEPENDENT CHECKS (f3i gates)		Verdict: NOVEL	
Eclipses observed	4 (gate >= 3)		
Robust depth ratio	1.09 (gate >= 0.50)		
Robust depth sigma	149.2 (gate >= 7.0)		
Odd/even z-score	6.65 (ALIAS FLAGGED)		
Secondary eclipse	not detected		
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)			
Sector 81	light-curve dip in this sector: 140.1 sigma	t0 refined by -0.0025 phase	
Difference-image peak SNR	74.28 (gate >= 5.0)	Compactness	0.92 (gate >= 0.50)
Dimming centroid offset	0.07 px (2 arcsec) (gate <= 1.0 px, i.e. on the target)		
Nearest bright-enough neighbour in cutout: TIC 349725132			
	0.57 px away, dTmag +0.87 - farther from the dimming than the target		



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 6.65 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 32.440640 d, in any write-up.

TIC 235512374

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 3.744060 d
Epoch (t0) 1472.0090 BTJD
Duration 0.0840 d
Depth 3.410 % (34,100 ppm)
SNR 173.8

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

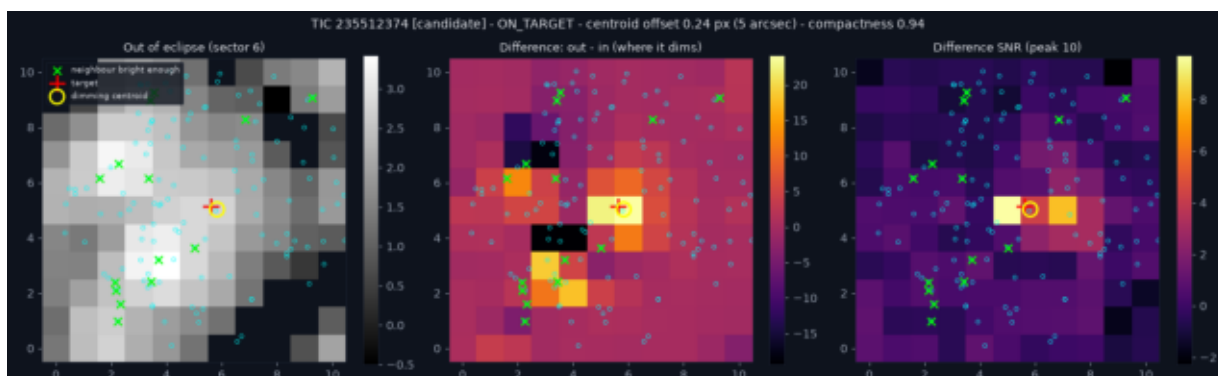
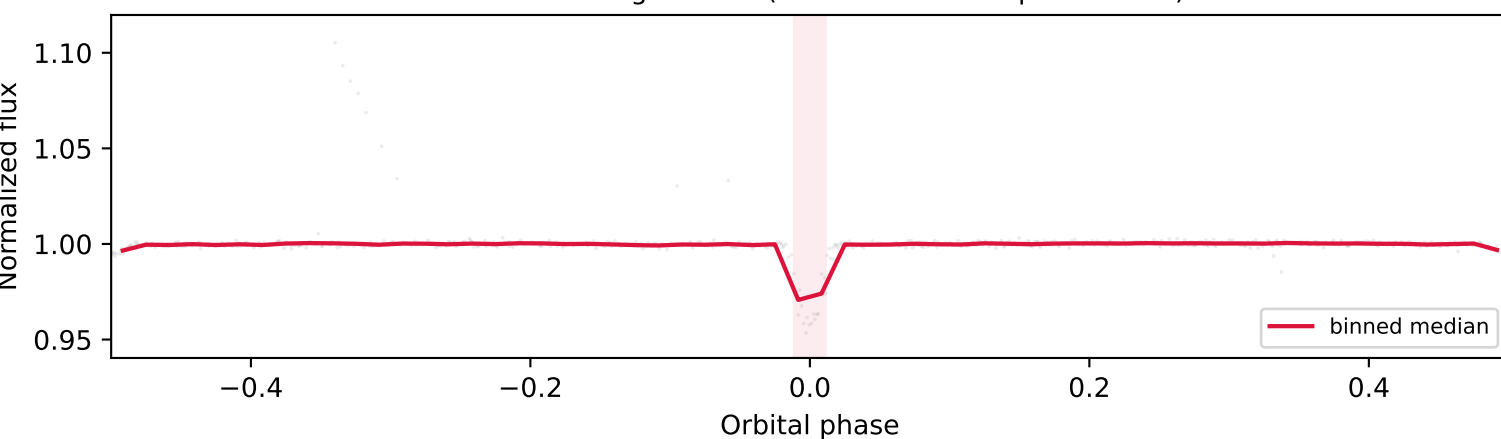
Eclipses observed 4 (gate ≥ 3)
Robust depth ratio 1.07 (gate ≥ 0.50)
Robust depth sigma 132.8 (gate ≥ 7.0)
Odd/even z-score 1.79 (consistent)
Secondary eclipse detected, phase -0.49, 6.6 sigma

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 6 light-curve dip in this sector: 119.5 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 9.71 (gate ≥ 5.0) Compactness 0.94 (gate ≥ 0.50)
Dimming centroid offset 0.24 px (5 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 235512361
1.60 px away, dTmag +2.72 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.

TIC 244999763

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 1.764116 d
Epoch (t0) 1598.3109 BTJD
Duration 0.1200 d
Depth 3.638 % (36,384 ppm)
SNR 159.8

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

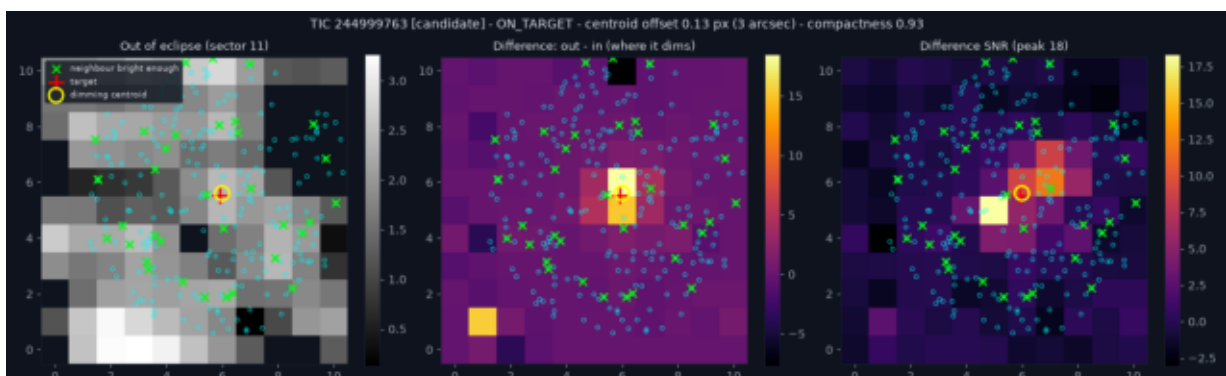
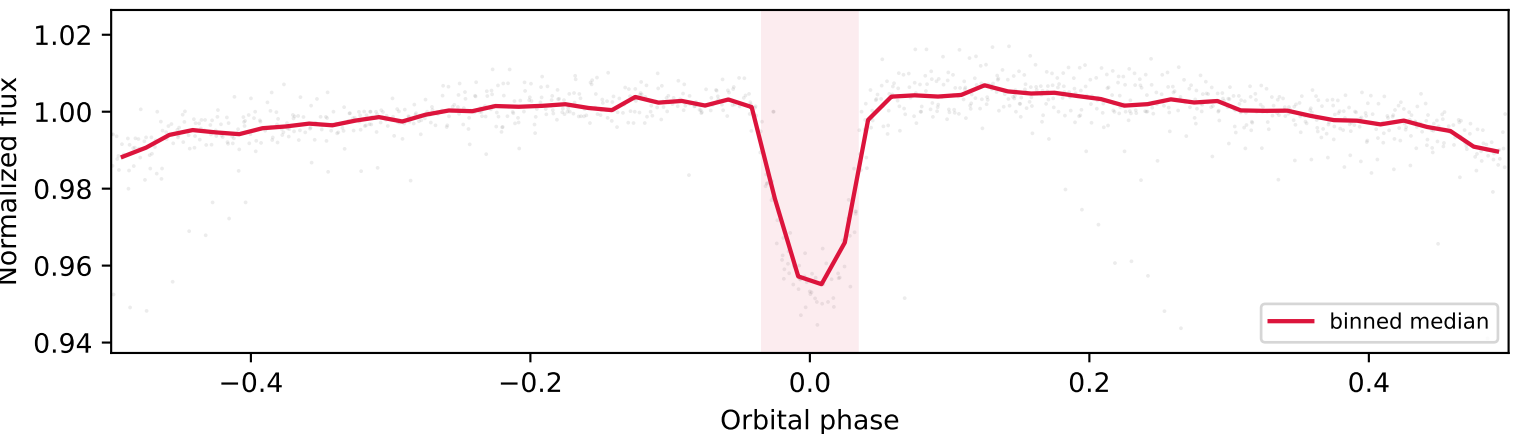
Eclipses observed 13 (gate ≥ 3)
Robust depth ratio 1.10 (gate ≥ 0.50)
Robust depth sigma 46.0 (gate ≥ 7.0)
Odd/even z-score 1.65 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 11 light-curve dip in this sector: 51.9 sigma t0 refined by 0.0075 phase
Difference-image peak SNR 18.35 (gate ≥ 5.0) Compactness 0.93 (gate ≥ 0.50)
Dimming centroid offset 0.13 px (3 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 244999756
0.58 px away, dTmag +2.53 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.

BLS EPHEMERIS		CATALOG CROSS-MATCH (novelty)	
Period	6.817344 d	VSX	no match
Epoch (t0)	1574.2365 BTJD	ASAS-SN	no match
Duration	0.5000 d	Gaia eclipsing-binary table	no match
Depth	1.909 % (19,085 ppm)	Gaia variable classification	no match
SNR	134.7	TESS EB catalog (Prsa+2022)	no match
INDEPENDENT CHECKS (f3i gates)		ExoFOP TOI list	no match
Eclipses observed	8 (gate >= 3)	Verdict:	NOVEL
Robust depth ratio	1.00 (gate >= 0.50)		
Robust depth sigma	43.2 (gate >= 7.0)		
Odd/even z-score	1.04 (consistent)		
Secondary eclipse	not detected		

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

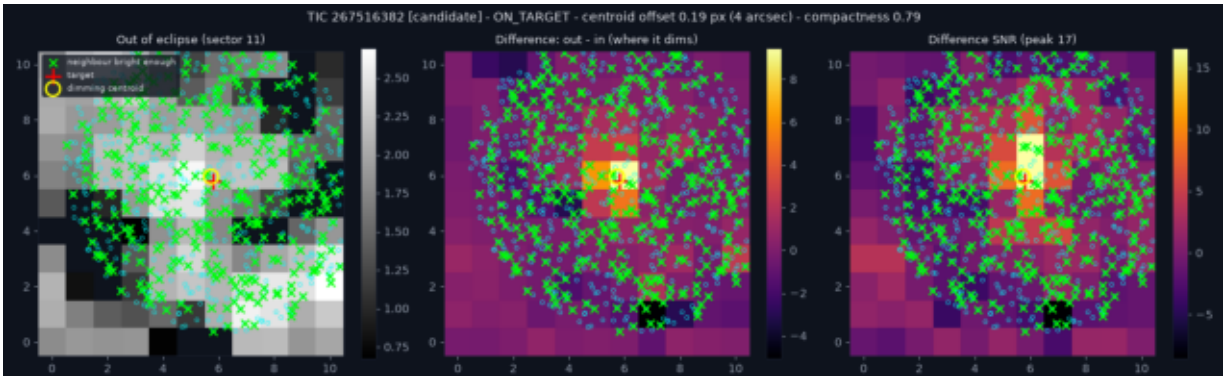
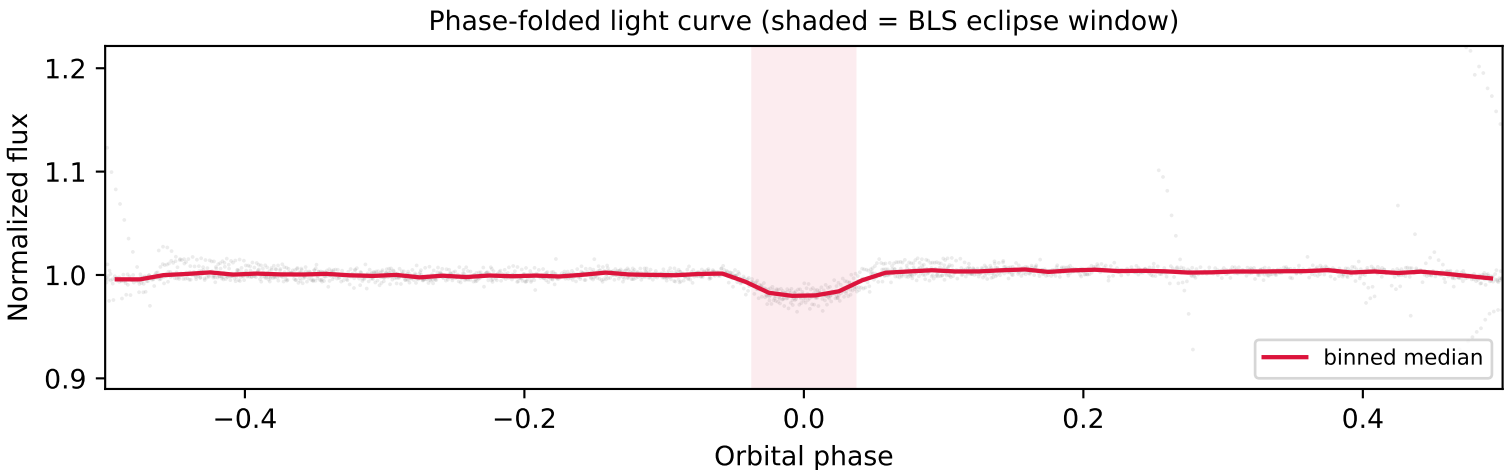
Sector 11 light-curve dip in this sector: 39.0 sigma t0 refined by -0.0025 phase

Difference-image peak SNR 16.62 (gate >= 5.0) Compactness 0.79 (gate >= 0.50)

Dimming centroid offset 0.19 px (4 arcsec) (gate <= 1.0 px, i.e. on the target)

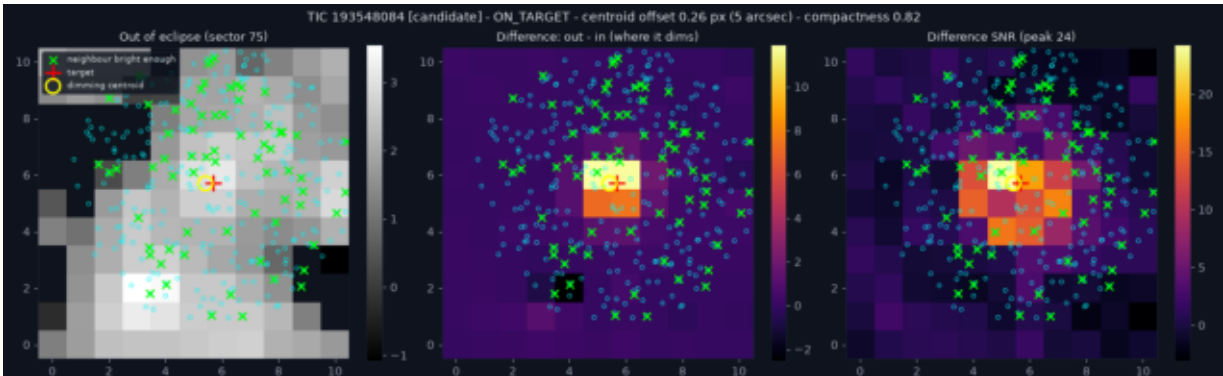
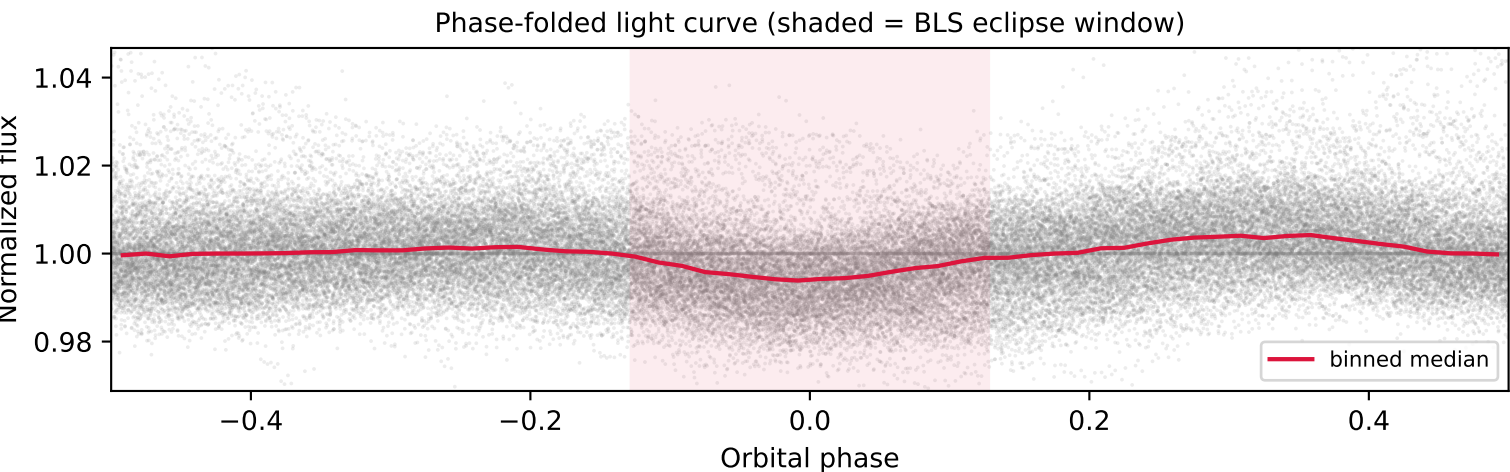
Nearest bright-enough neighbour in cutout: TIC 267516364

0.21 px away, dTmag +2.34 - farther from the dimming than the target



Caveat (THIN MARGIN): The ON_TARGET call rests on a neighbour-competition margin of just 0.020 px (dimming centroid 0.191 px from target vs 0.211 px from the nearest bright-enough neighbour) - inside diag_pixel.py's own stated centroid systematic of 0.10-0.3 px. TESS's resolution cannot confidently separate this star from that neighbour.

BLS EPHEMERIS		CATALOG CROSS-MATCH (novelty)	
Period	0.953043 d (ALIASED, see below)		
-> corrected period: 1.906087 d		VSX	no match
Epoch (t0)	1683.5509 BTJD	ASAS-SN	no match
Duration	0.2440 d	Gaia eclipsing-binary table	no match
Depth	0.517 % (5,171 ppm)	Gaia variable classification	no match
SNR	83.1	TESS EB catalog (Prsa+2022)	no match
		ExoFOP TOI list	no match
INDEPENDENT CHECKS (f3i gates)		Verdict: NOVEL	
Eclipses observed	252 (gate >= 3)		
Robust depth ratio	0.97 (gate >= 0.50)		
Robust depth sigma	61.4 (gate >= 7.0)		
Odd/even z-score	3.39 (ALIAS FLAGGED)		
Secondary eclipse	not detected		
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)			
Sector 75	light-curve dip in this sector: 48.3 sigma	t0 refined by 0.0225 phase	
Difference-image peak SNR	24.25 (gate >= 5.0)	Compactness 0.82 (gate >= 0.50)	
Dimming centroid offset	0.26 px (5 arcsec)	(gate <= 1.0 px, i.e. on the target)	
Nearest bright-enough neighbour in cutout: TIC 193548086			
0.59 px away, dTmag +4.81 - farther from the dimming than the target			



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 3.39 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 1.906087 d, in any write-up.

TIC 133688675

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 2.852426 d
Epoch (t0) 1492.4576 BTJD
Duration 0.1200 d
Depth 1.951 % (19,508 ppm)
SNR 71.7

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

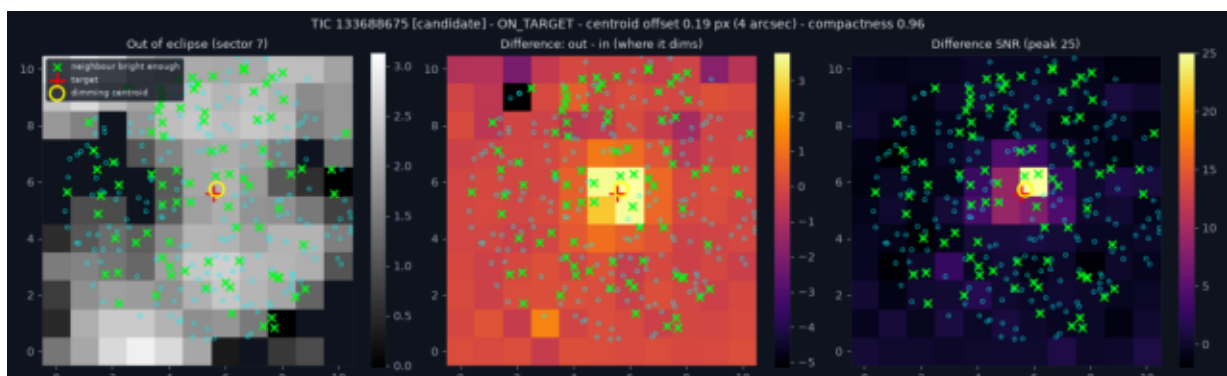
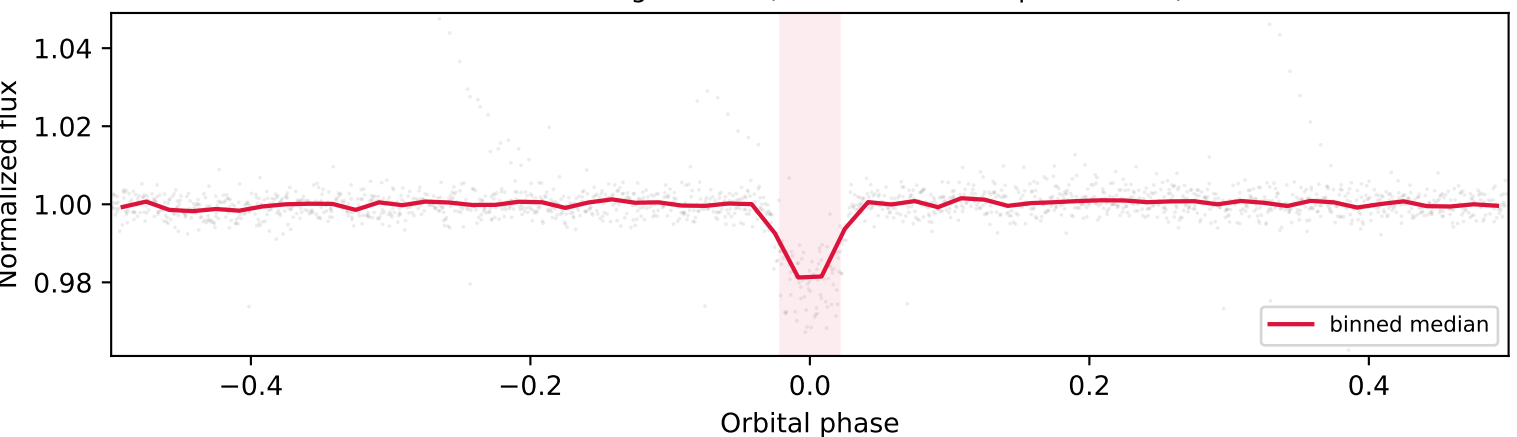
Eclipses observed 14 (gate ≥ 3)
Robust depth ratio 0.96 (gate ≥ 0.50)
Robust depth sigma 48.3 (gate ≥ 7.0)
Odd/even z-score 0.37 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 7 light-curve dip in this sector: 33.0 sigma t0 refined by 0.0025 phase
Difference-image peak SNR 25.09 (gate ≥ 5.0) Compactness 0.96 (gate ≥ 0.50)
Dimming centroid offset 0.19 px (4 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 133688658
0.49 px away, dTmag +1.39 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.

TIC 224215371

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 5.493091 d (ALIASED, see below)
-> corrected period: 10.986182 d
Epoch (t0) 2449.2516 BTJD
Duration 0.0840 d
Depth 1.431 % (14,311 ppm)
SNR 52.2

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

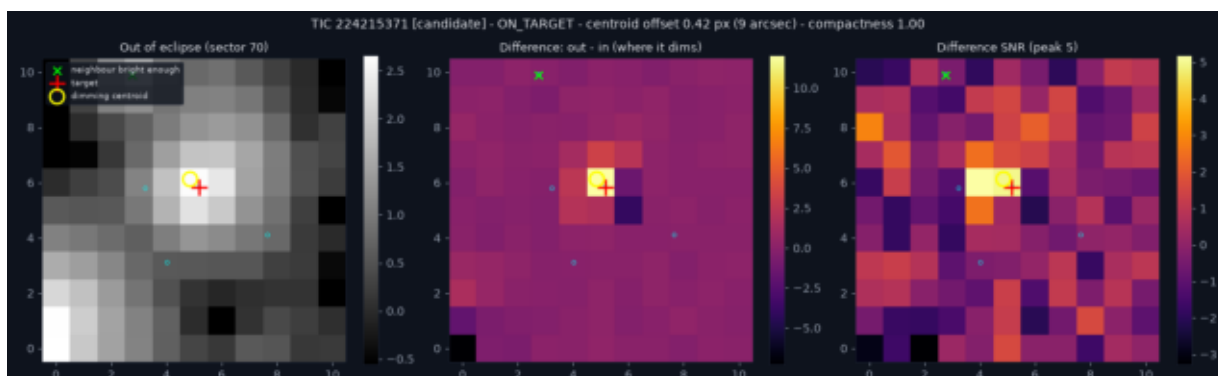
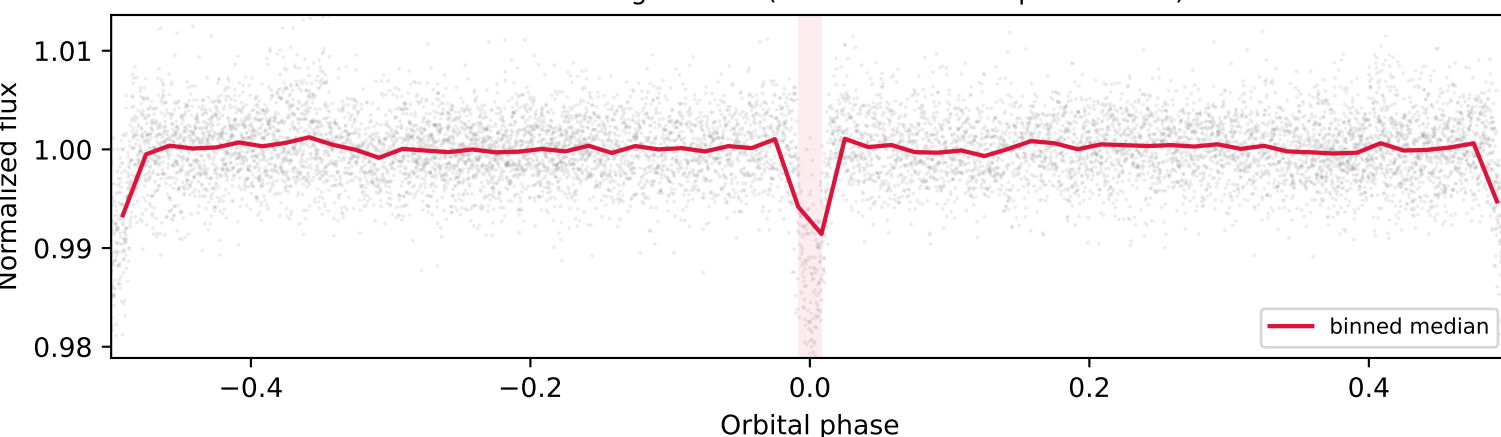
Eclipses observed 5 (gate ≥ 3)
Robust depth ratio 0.88 (gate ≥ 0.50)
Robust depth sigma 34.5 (gate ≥ 7.0)
Odd/even z-score 6.36 (ALIAS FLAGGED)
Secondary eclipse detected, phase 0.49, 6.7 sigma

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 70 light-curve dip in this sector: 28.8 sigma t0 refined by 0.0025 phase
Difference-image peak SNR 5.18 (gate ≥ 5.0) Compactness 1.00 (gate ≥ 0.50)
Dimming centroid offset 0.42 px (9 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 224217503
4.32 px away, dTmag +2.56 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 6.36 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 10.986182 d, in any write-up.

TIC 139805341

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 2.931491 d
Epoch (t0) 1327.0564 BTJD
Duration 0.0840 d
Depth 1.083 % (10,834 ppm)
SNR 22.6

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

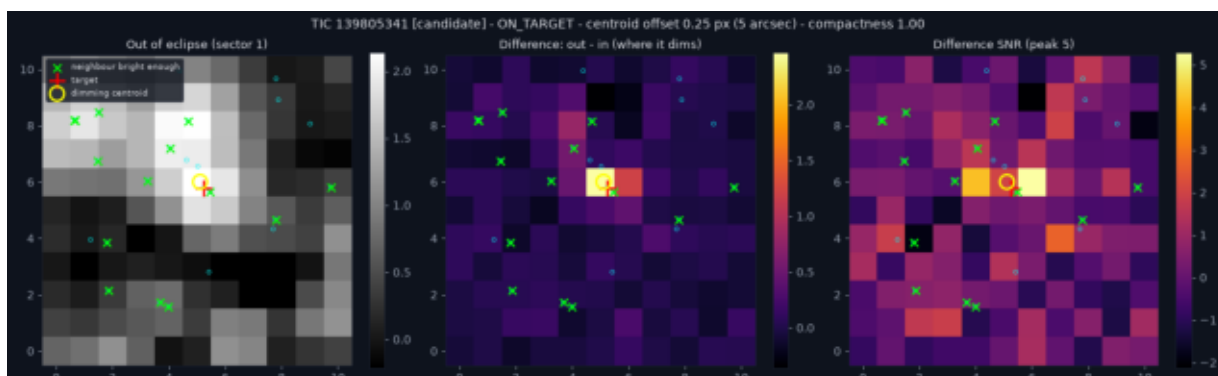
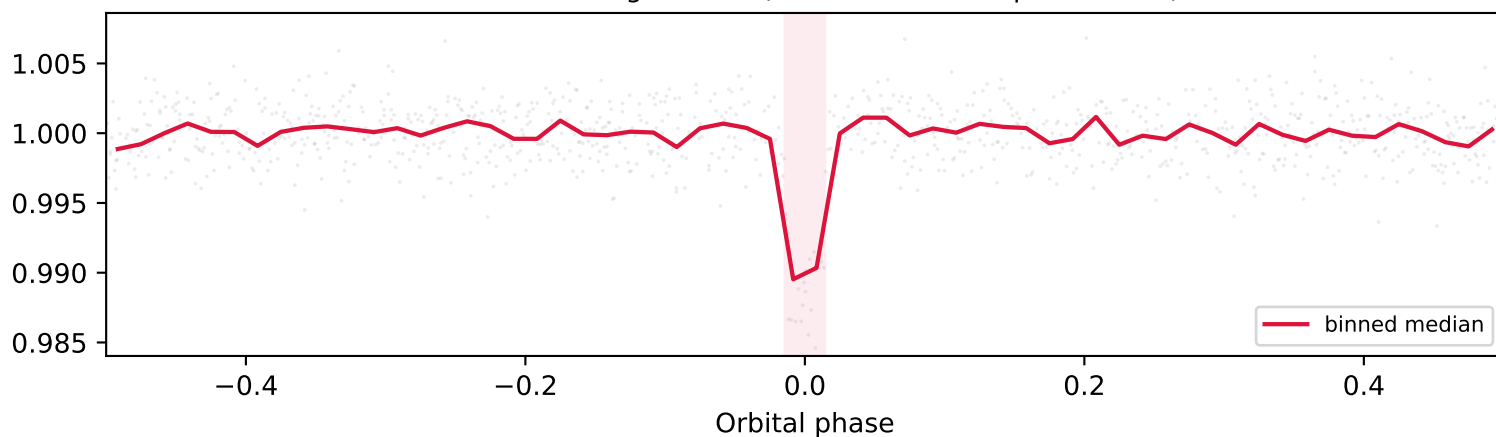
Eclipses observed 4 (gate ≥ 3)
Robust depth ratio 0.97 (gate ≥ 0.50)
Robust depth sigma 20.4 (gate ≥ 7.0)
Odd/even z-score 2.88 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 1 light-curve dip in this sector: 19.8 sigma t0 refined by -0.0025 phase
Difference-image peak SNR 5.29 (gate ≥ 5.0) Compactness 1.00 (gate ≥ 0.50)
Dimming centroid offset 0.25 px (5 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 139805343
0.50 px away, dTmag +3.85 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



TIC 43330648

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 14.285582 d
Epoch (t0) 1991.2750 BTJD
Duration 0.1700 d
Depth 0.031 % (312 ppm)
SNR 15.1

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

Eclipses observed 6 (gate >= 3)
Robust depth ratio 0.94 (gate >= 0.50)
Robust depth sigma 10.9 (gate >= 7.0)
Odd/even z-score 0.62 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 26 light-curve dip in this sector: 6.5 sigma t0 refined by 0.1325 phase
Difference-image peak SNR 25.82 (gate >= 5.0) Compactness 0.84 (gate >= 0.50)
Dimming centroid offset 0.79 px (17 arcsec) (gate <= 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 43330659
2.01 px away, dTmag +7.83 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)

